FrutaMIDI: capacitive USB MIDI controller

Engineering report

FrutaMIDI engineering project

FrutaMIDI v1.2 (change request CR-002), interfaces frozen-6, requirements draft-8, 2026-11-07

Prepared for release V1.2 (git tag `v1.2.0`); not yet released. Previous release: V1_BASELINE (git tag `v1.1.0-baseline`)

Abstract

FrutaMIDI is a USB MIDI controller that sends notes when a user touches conductive objects wired to a Raspberry Pi Pico and an Adafruit MPR121 capacitive sensor module, built on a breadboard or on a single-sided home-etched PCB (v1.2). This report collects the requirements, decisions, interfaces, analyses and host tests for FrutaMIDI v1.2. Version 1.2 (change request CR-002) replaces the electrode series resistors by pulse-rated parts lying flat. By analysis of the design files, a 4 kV contact discharge should then neither arc across a resistor nor exceed its pulse rating; this has not been tested. A manager PCB design review then widened the JP1 solder-jumper gaps, enlarged the screw-terminal pads, tied the potentiometer tabs to AGND and added a 100 nF wiper filter (ADR-021). Section 12 lists the changes since the V1_BASELINE release. Every number is copied from a cited project file with its label. **No part of the hardware has been built or measured**: analyses, simulations, design-rule checks and host tests are not hardware validation, and the physical validation plan in Section 10 lists what remains to be measured.

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1 Introduction and scope

FrutaMIDI is a USB MIDI controller that turns touches on conductive objects (fruit, plants, water) into MIDI notes. It uses a Raspberry Pi Pico board and an Adafruit MPR121 breakout (product 1982). The project continues an existing repository (v1.0.0 released) and covers two builds: a solderless breadboard and a single-sided PCB for home etching, now at label "FrutaMIDI v1.2" [`management/DECISIONS.md`] (ADR-000; ADR-019 D-2 supersedes the "v1.1" label of ADR-008). Version 1.2 is the result of change request CR-002 [`changes/CR-002.md`]; the changes since the V1_BASELINE release are listed in Section 12.

1.1 Scope

- In scope: electronics design (KiCad, generated by scripts), Pico firmware, the browser sampler (`docs/index.html`), host tests, user documentation and this report [`CLAUDE.md`].
- Out of scope: ordering parts, flashing or powering hardware, and any physical measurement. Hence every [PHYSICAL] requirement is at most PARTIAL or NOT RUN [`management/REQUIREMENTS.md`].
- Hosts: Windows 10, Windows 11 and macOS 12 or later (ADR-007 D-6).

1.2 Document conventions

- Every number carries one label: SOURCE, CALCULATION, SIMULATION, MEASUREMENT or ASSUMPTION, and a citation in brackets to the repository file it is copied from.
- Status is reported only as PASS, PARTIAL, FAIL, BLOCKED or NOT RUN.
- Passing ERC, DRC or a compile is not hardware validation.

2 Requirements and verification status

Source: [`management/REQUIREMENTS.md`], revision draft-8 (T-077, CR-002), proposed and not yet baselined. Draft-6 updated the evidence of REQ-023, REQ-028, REQ-029 and REQ-032 for the pulse-rated R1–R12; draft-7 (T-074) refreshed the evidence after the re-review fixes; draft-8 updates the evidence of REQ-006, REQ-029, REQ-032 and REQ-036 for the manager PCB review follow-up (ADR-021). No status changed. Methods: A analysis, S simulation, I inspection, T test. By the project rule, a requirement whose analysis or inspection passes while its physical test is NOT RUN is PARTIAL (ADR-009).

Summary. Of 42 requirements (counts unchanged by CR-002): PASS 17, PARTIAL 18, FAIL 3, NOT RUN 4 [`management/REQUIREMENTS.md`].

- FAIL: REQ-018 VBUS range (the USB low-power-port minimum of 4.40 V is below the Pico's recommended 4.50 V, SOURCE S-01, S-10); REQ-032 core BOM cost, 23.98–38.62 EUR against $\leq$ 30 EUR (CALCULATION, budgets revision 7; nearly all prices ASSUMPTION), upper bound over the target by 8.62 EUR (was 0.40 EUR before CR-002); the R1–R12 increase is accepted for ESD safety (ADR-019 D-3) and the C1 increase (0.20–0.90 EUR, ASSUMPTION) by the ADR-021 addendum, the status stays FAIL; REQ-040 JP1 wrong-position tolerance, kept FAIL with accepted residual risk R-007 (ADR-013) [`requirements/requirements.csv`].
- NOT RUN (physical only): REQ-010 status LED, REQ-011 USB MIDI class compliance, REQ-037 sensing range, REQ-038 false triggers and crosstalk.

The table below is generated from `requirements/requirements.csv` by `reports/report/scripts/csv_to_table.py`; do not edit it by hand.

Table 1: Requirements and verification status (generated from requirements/requirements.csv).

ID	Requirement	Verification	Status	Evidence
REQ-001	**Note On/Off per electrode** Each of the 12 electrode inputs E0–E11 (MPR121 ELE0–ELE11) shall send one MIDI Note On, on the configured channel (default 1) with the configured fixed velocity (default 100, range 1–127), when a touch is detected. It shall send a Note Off for the same note when the touch is released.	I + T [PHYSICAL]	PARTIAL	I: PASS. The chain terminal → series resistor → module electrode is 1:1 (interface_ check `pcb.A2.pin2..13`, `pcb.J2.*`). T: NOT RUN.
REQ-002	**Note Off integrity** The Note Off for a pad shall carry the note number sent in its Note On, even if the scale or octave changed in between. All sounding notes shall be released before a recalibration.	T (host unit test) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Hardware test NOT RUN.
REQ-003	**Second sensor (24 pads)** The firmware shall support a second MPR121 at I2C address 0x5B (`NUM_SENSORS = 2`), mapping its electrodes to pads 12–23. This is a breadboard-only option; the PCB carries one module (ADR-007 D-3).	A + T [PHYSICAL]	PARTIAL	A: latency PASS at 400 kHz (12.35 ms $\leq$ 15 ms, jitter 8.35 ms), using the worst-case run-based burst plan (budgets.out §b). T: NOT RUN.
REQ-004	**Polyphonic aftertouch** While a pad is touched, the device shall send Polyphonic Key Pressure (0xAn) with a value 0–127. The value shall map linearly from delta = baseline − filtered, clamped to [12, 120]. It shall be evaluated every 15 ms and sent only when it changes by at least 2.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Aftertouch messages are sent after all Note On/Off of the same loop (ADR-010). Hardware NOT RUN.
REQ-005	**Six scales** The device shall provide 6 scales, selected cyclically: Drum Rack (36–47, octave ignored, the power-on default), major pentatonic, minor pentatonic, major, natural minor and chromatic (root 48). Notes shall be clamped to 0–127.	A + T (host)	PASS	A: notes 24–99 with 12 pads and 24–127 with 24 pads, never clamped (T-009 budgets §d). Host tests PASS (T-040 `test_scales.py`).

ID	Requirement	Verification	Status	Evidence
REQ-006	**Octave potentiometer** The octave potentiometer shall shift melodic scales by −2..+2 octaves using 5 ADC zones, each V33/5 = 0.660 V wide. The octave shall change only when the reading is more than 0.7 zone from the current zone centre, i.e. 0.2 zone (0.132 V) beyond the border, which is the dead band. The combined ADC noise, DNL, quantisation and supply ripple shall stay below the dead band. A static ADC offset shifts every threshold together and does not consume the dead band. The pot shall have no effect on Drum Rack and shall be enabled by default on the PCB build.	A + T (host) + T [PHYSICAL]	PARTIAL	A: PASS. Error budget 0.0364 V < dead band 0.132 V, margin 0.096 V (budgets A5). Inputs: noise from ENOB 8.7 (SOURCE S-17 Table 627); DNL peak 10 LSB12 (ASSUMPTION; S-17 §4.9.4 and erratum RP2040-E11 give peaks at codes 512/1536/ 2560/3584, height only in a figure); 10-bit step; ripple 10 mV (ASSUMPTION). The nearest DNL-peak code is 0.049 V from a switch point. The static offset of about 30 mV (S-01 §4.3) is reported, not budgeted (review F-003). Host tests PASS (T-043). C1 wiper filter (ADR-021, T-077): 100 nF X7R KEMET C322C104K5R5TA from POT to AGND; worst Thevenin source 3.0 kOhm (10 kOhm +20 % ASSUMPTION, mid travel) and worst C 126.5 nF (+10 % tolerance, +15 % X7R temperature, SOURCE EE-S29) give tau 0.380 ms (nominal 0.25 ms) and settling to 1/2 LSB at 10 bits in 2.89 ms against the 30 ms read period (SOURCE fm_ pot.h POT_READ_MS), margin 10.4x (CALCULATION, budgets A7 PASS, criterion 5x ASSUMPTION; agrees with sim/electronics/calc/calc_ pot_rc.out). C1 does not enter the error budget, which keeps the 10 mV ripple ASSUMPTION. Hardware NOT RUN.
REQ-007	**Button short press** A short press of the button (released within 800 ms, sampled every 10 ms) shall select the next scale, and the LED shall blink N times for scale N.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Hardware NOT RUN.
REQ-008	**Button long press recalibration** A press of more than 800 ms shall release all notes and recalibrate all sensors; the LED then blinks 3 times.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Recalibration blocks for 16.8–22.1 ms per sensor at 400 kHz (CALCULATION); this is user-requested and allowed by REQ-016. Hardware NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-009	**Sensor absent or lost** If no MPR121 answers at start, the device shall send no MIDI from it and retry. During play, a sensor shall be marked lost after 3 consecutive failed I2C transactions, or at once when its ECR read-back (every 250 ms) differs from the value read after start-up (a chip that reset itself). On loss the firmware shall release that sensor's notes, show the fast error blink and stop reading it. It shall retry start-up when more than 2 s have passed since the last start attempt (this also covers a second sensor missing at boot). The other sensor shall keep playing.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS: `tests/firmware/logic/test_notes_health.py` (15 tests), within `295 passed` (T-044 handoff). Implementation: `fm_notes.h`/ health logic, `startSensor()` = begin → setClock(400000) → configure → ECR read. A self-reset chip releases its notes through the normal touch path before the ECR check, so only one Note Off is sent. Physical NOT RUN (it needs the sensor unplugged or reset during play).
REQ-010	**Status LED** The LED shall give 2 slow blinks at start when a sensor is found, and shall be lit while any pad is touched. On Pico W and Pico 2 W the LED is driven through the CYW43 (WL_GPIO0).	T [PHYSICAL]	NOT RUN	
REQ-011	**USB MIDI class compliance** The device shall enumerate as a USB MIDI 1.0 class device (MIDIStreaming, bulk endpoints). It shall work without vendor drivers on Windows 10, Windows 11 and macOS 12 or later.	T [PHYSICAL]	NOT RUN	
REQ-012	**Device name** The USB product string and the MIDI port name shall be "FrutaMIDI".	I + T [PHYSICAL]	PARTIAL	I: `DEVICE_NAME = "FrutaMIDI"` is passed to `setProductDescriptor` and `setStringDescriptor`. T: NOT RUN.
REQ-013	**MIDI message set** The device shall send only Note On (0x9n, CIN 0x9), Note Off (0x8n, velocity 0, CIN 0x8) and Polyphonic Key Pressure (0xAn, CIN 0xA), on one channel n = MIDI_CHANNEL − 1. Incoming MIDI shall be read and discarded.	I	PASS	`fm_midi.h` constants match S-14 Table 4-1.
REQ-014	**Touch-to-MIDI latency** Device latency from the first touch after a pad has been idle for at least 15 ms (DEBOUNCE_MS) to the USB-MIDI packet being available to the host shall be $\leq$ 15 ms worst case, for electrode capacitance $\leq$ 100 pF, with mitigation M1 (one burst read per sensor per aftertouch pass). A re-touch of the same pad within the 15 ms lockout is outside this requirement and is reported separately.	A + T [PHYSICAL]	PARTIAL	A: PASS. 1 sensor 10.72 ms, 2 sensors 12.35 ms at 400 kHz, including the T-044 ECR poll ($\approx$ 0.14 ms per sensor) (CALCULATION, budgets.out §b). Separate term: a re-touch inside the lockout adds up to 15 ms, giving 25.72 ms worst case (review F-011). Design limit C $\leq$ 210 pF. T: NOT RUN (plan item P-2). Flagged for user review (ADR-007).
REQ-015	**Latency jitter** Latency jitter (worst minus best) shall be $\leq$ 10 ms for electrode capacitance $\leq$ 100 pF.	A + T [PHYSICAL]	PARTIAL	A: PASS. 6.72 ms (1 sensor) and 8.35 ms (2 sensors) at 400 kHz, including the ECR poll (CALCULATION). T: NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-016	**No blocking during play** During play, no firmware action other than a user-requested recalibration or the automatic restart of a lost sensor shall block touch processing for more than 10 ms.	I + A	PASS	I: only delay(10) in midiBegin() (setup); non-blocking LED (fm_ blink.h). A: worst loop ~1.72 ms, 1 sensor, 400 kHz (sim/ systems/budgets.out, CALCULATION). Exception: lost-sensor restart 16.8-22.1 ms per retry, >=2 s apart (review N-001, ADR-018).
REQ-017	**USB bus power only** The device shall be powered only from USB VBUS through the Pico's micro-USB connector. No other power input shall exist.	I	PASS	Pico pins 39 and 40 are unconnected (`pcb.A1.no_ other_nets`).
REQ-018	**VBUS range** The device shall operate with VBUS from 4.40 V to 5.50 V at its connector.	A + T [PHYSI-CAL]	FAIL	A: VSYS 3.95–5.25 V, inside the 1.8–5.5 V range (S-01 §2.1): PASS. However, the 4.40 V low-power-port minimum (S-10) is below the Pico's recommended VBUS of 5 V ± 10 % = 4.50 V (S-01 §2.3): budgets check A3b FAIL, so the requirement as stated FAILs (review F-010). Electrically the Pico still runs (VSYS margin 2.15 V). Options: restrict the statement to 4.50–5.50 V, or accept as a documented deviation. T: NOT RUN.
REQ-019	**VBUS current** VBUS current shall be ≤ 100 mA (the declared bMaxPower, S-11) for every supported Pico board.	A + T [PHYSI-CAL]	PARTIAL	A: worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W. Lower bounds are SOURCE (S-01 Table 3, S-16 Table 2); RP2040 pull-ups 50–80 kΩ are SOURCE S-17. The upper bounds are ASSUMPTIONs. T: NOT RUN (plan item P-3).
REQ-020	**3V3 rail load** The external load on the Pico 3V3(OUT) shall be ≤ 300 mA with at least 50 % margin.	A	PASS	10.15 mA worst case with 2 sensors and PCB pull-ups: 96.6 % margin.
REQ-021	**MPR121 supply voltage** The MPR121 VDD, fed from the Pico 3V3 through the module LDO, shall be within 2.0–3.6 V.	A	PASS	3.10–3.40 V (CALCULATION; dropout and rail tolerance are ASSUMPTIONs).
REQ-022	**Touchable voltage (SELV)** Every user-touchable conductor shall be at ≤ 3.6 V relative to GND and shall be supplied only from the USB SELV source.	A	PASS	Electrodes are charged from MPR121 VDD ≤ 3.6 V (S-04), and REQ-017 holds.

ID	Requirement	Verification	Status	Evidence
REQ-023	**Electrode ESD robustness** Each user-touched electrode terminal shall survive ±4 kV contact discharge without permanent damage. The ±4 kV target is an ASSUMPTION, following IEC 61000-4-2 level 2 (the standard was not read).	I + A + T [PHYSICAL]	PARTIAL	Estimate on ASSUMPTIONs, not a PASS (review F-002). Current: with 3.3 kΩ (R1–R12) plus the IEC 330 Ω source, the peak at 4 kV is 4 kV / 3.63 kΩ = 1.10 A, below the 1.33 A pin limit inferred from the HBM ±2 kV rating (the limit is itself an ASSUMPTION); IEC 4 kV through 3.63 kΩ gives τ = 545 ns and Q = 600 nC against 150 ns / 200 nC for HBM 2 kV (CALCULATION). Voltage across R: 3.64 kV at 4 kV, 7.27 kV at 8 kV (CALCULATION). Pulse rating (CR-002, ADR-019 D-1): R1–R12 are Vishay MBE04140C3301FC100 (DIN0414, 3k3 1 %), 4.08 kV 1.2/50 µs at 3.3 kΩ (SOURCE EE-S17b chart; ±5 % reading ASSUMPTION), margin 1.12 at 4 kV (1.07 at −5 %) (CALCULATION); rating-pulse energy 182 mJ from (1/R)integral of V^2dt, chart energy-limited below the 5073 Ω knee, plateau 5054 V (CALCULATION, C2-003, sim/electronics/calc/calc_pulse_rated.out); verdict PASS at 4 kV, FAIL at 8 kV. Flashover around R (gap rule ≥ 3.5 mm at 4 kV, ≥ 7.5 mm at 8 kV; CALCULATION, breakdown sweep on ASSUMPTION models, rule from model M4; IEC 60664-1 F.2 values recalled, EE-S25), sim/electronics/calc/check_gap.out on the c901ad8 design: (a) copper ELEx–Ex 13.04 mm and (b) Ex to any other ELE/MPR121/Pico net 10.92 mm on all 12 channels (MEASUREMENT on the design file): PASS at 4 kV and 8 kV (copper only). (d) end caps, same channel and cross-channel (CALCULATION on ASSUMPTIONS: body L 11.9 mm SOURCE EE-S27 / 11.0 mm ASSUMPTION, cap length Lc 1.5–3.5 mm ASSUMPTION, placement error 0 / ±0.25 / ±0.5 mm ASSUMPTION): worst 3.61 mm at ±0.25 mm, PASS at 4 kV; worst 3.13 mm at ±0.5 mm, FAIL; the 4 kV rule holds at every swept L and placement error if Lc ≤ 3.0 mm; FAIL at 8 kV in every case. End caps are therefore PARTIAL at 4 kV; ADR-020 controls this by forming the

ID	Requirement	Verification	Status	Evidence
REQ-024	**I2C hand-wire fault current** A misplaced SDA or SCL hand wire (including one on a supply hole S1/S2/S6/S7) shall not drive an RP2040 pad beyond its IO current limits.	I + A	PASS	I: 220 Ω series resistors are present between the Pico pads and the bus: pad 31 SDA_PICO → R16 → SDA, pad 32 SCL_PICO → R15 → SCL (interface_check `fw.pcb.PIN_SDA/SCL_vs_pcb`). A: fault current 15.5 mA per line at 3.4 V, 30.9 mA for both lines, ≤ IIOVDD_MAX 50 mA (SOURCE S-17 Table 625). Low-level margins: Pico→MPR121 0.138 V, MPR121→Pico 0.235 V (budgets B5/B6, CALCULATION). Statement reworded after ADR-009 D-9: a wire on a supply hole is still physically possible, but it is now current-limited.
REQ-025	**Both module revisions** One PCB shall accept both Adafruit 1982 revisions. JP1 bridges ORIG (+3V3 → S1 = VIN) or QT (+3V3 → S7 = VCC). The SDA/SCL wires from J1 go to S5/S6 (original) or S3/S4 (STEMMA QT).	I + T [PHYSICAL]	PARTIAL	I: PASS. JP1 selector S1/+3V3/S7 with silkscreen legend 'ORIG = no white connectors' and 'QT = 2 white connectors' (`pcb.JP1_legend`). J1 1 = SCL, 2 = SDA. hardware/README.md table (JP1, SDA, SCL) = ORIG/S5/S6 and QT/S3/S4, which matches the Eagle-derived S-row (`doc.hw_readme.jp1_and_wires.*`). T: NOT RUN.
REQ-026	**I2C rise time (PCB)** The PCB-build I2C bus shall meet the rise time for its clock: ≤ 300 ns at 400 kHz (≤ 1000 ns at 100 kHz), with the module at 0x5A.	A + T [PHYSICAL]	PARTIAL	A: PASS. R13/R14 4.7 kΩ: t_r 156 ns at 60 pF, C_b limit 115.2 pF (systems, with RPU 80 kΩ from S-17); electronics gives 103 ns at 38 pF. The 220 Ω series resistors add 2.2 ns (ASSUMPTION of 10 pF pad capacitance). IO levels with 220 Ω: B5–B8 PASS, minimum margin 0.138 V (S-17 Table 625, S-04 Table 5). T: NOT RUN (plan item P-4).
REQ-027	**Single-sided PCB** The PCB shall be single-sided: copper on B.Cu only, through-hole parts only, no vias, no copper pour, and no wire bridges apart from the 2 I2C configuration wires and the JP1 solder bridge.	I	PASS	pcb_rules_check: all segments on B.Cu; 0 vias, 0 zones, 0 SMD pads.

ID	Requirement	Verification	Status	Evidence
REQ-028	**Board outline** The board outline shall be 100 × 70 mm.	I	PASS	Outline 100.000 × 70.000 mm (sim/systems/pcb_rules_check.out, re-run on c901ad8, T-074: `PASS REQ-028 board outline 100 x 70 mm :: outline 100.000 x 70.000 mm`). Board label 'FrutaMIDI v1.2' (ADR-019 D-2): interface_check `PASS pcb.label_v1.2` on c901ad8 (the earlier FAIL on 3a3684b is closed). The label is not part of this requirement's statement.
REQ-029	**Home-etch rules** Home-etch rules: signal tracks ≥ 1.0 mm (≥ 0.8 mm for short stubs), power tracks ≥ 1.2 mm, clearance ≥ 0.6 mm (DRC hard minimum 0.5 mm), annular ring ≥ 0.4 mm, drills ≥ 0.8 mm.	I + A (DRC)	PASS	sim/systems/pcb_rules_check.out re-run on a216c1b (T-077): `SUMMARY: 19 checks, 19 PASS, 0 FAIL`. The annular-ring check is now offset-aware (J2 drill offset 0.75 mm: ring 0.80 mm). New checks for ADR-021: `ADR-021 JP1 pad gap >= 1.8 mm` PASS (SolderJumper_3_THT_P3.81mm, pitch 3.81 mm, gaps 1.87/1.87 mm) and `ADR-021 J2 pads 3.0 x 4.5 mm` PASS (14 pads). C1 footprint C_Radial_KEMET_C322_P5.08mm_HomeEtch: ring 0.60 mm, drill 0.8 mm. R_Axial_DIN0414_P15.24mm_Horizontal_HomeEtch (R1–R12): drill 1.1 mm, ring 0.55 mm (C2-008). Minimum annular ring on the board 0.40 mm; minimum drill 0.8 mm. Both new checks were mutation-tested (JP1 pitch 3.6 mm gives FAIL at 1.66 mm). Clearance relies on the electronics DRC (reports/electronics/evidence/T-076/drc.rpt: `Found 0 DRC violations`, per the T-076 handoff), which systems did not re-run.
REQ-030	**Screw-terminal map** The PCB shall provide 14 screw terminals at 5.08 mm pitch: GND, E0..E11, GND. Terminal k (k = 2..13) shall reach MPR121 electrode k − 2 through its series resistor.	I	PASS	interface_check J2.T1..T14 and A2.pin2..13.
REQ-031	**Pin map consistency** The firmware PCB pin map (I2C1 SDA GP26 / SCL GP27, button GP22, pot GP28/ADC2) shall match the PCB nets. The breadboard map (I2C0 GP4/GP5, button GP15, pot GP26) shall match README.md.	I	PASS	interface_check: 0 pin mismatches.

ID	Requirement	Verification	Status	Evidence
REQ-032	**Core BOM cost** The core BOM (assembled board, excluding tools, consumables and USB cable) shall cost $\leq$ 30 EUR.	A	FAIL	23.98–38.62 EUR (CALCULATION, sim/systems/budgets.out revision 7, T-077). New row C1 100 nF KEMET C322C104K5R5TA: 0.20–0.90 EUR (ASSUMPTION from a distributor search, EE-S30, prices not verified; ADR-021), adding 0.20/0.90 EUR to the lower/upper bound. R1–R12 pulse-rated Vishay MBE04140C3301FC100: 12 × 0.30–0.66 = 3.60–7.92 EUR (SOURCE EE-S26 sibling-value listings; equal 3k3 price ASSUMPTION). The upper bound exceeds the target by 8.62 EUR (was 7.72 EUR); the lower bound meets it. The R1–R12 increase is accepted for ESD safety (ADR-019 D-3); the C1 increase is not yet covered by an acceptance; the status stays FAIL. Most other prices are ASSUMPTIONs; quoted prices from one supplier would narrow the range.
REQ-033	**Firmware build matrix** The firmware shall build without errors for Pico, Pico W, Pico 2 and Pico 2 W, each for the breadboard and PCB variants (FQBN `rp2040:rp2040:<board>:usbstack=tinyusb`, plus `-DFRUTAMIDI_PCB` for PCB), and for `arduino:avr:leonardo` (optional).	T (compile)	PASS	`reports/firmware/T-044/build/*.log`: 9 of 9 logs (Pico, Pico W, Pico 2, Pico 2 W × std/PCB, plus Leonardo) contain 'Sketch uses'. A clean compile is not a hardware validation.
REQ-034	**Browser sampler** The browser sampler `docs/index.html` shall work in Chrome and Edge (Web MIDI) on Windows and macOS. It shall map its 12 pads to notes 36–47 by default, treat Note On with velocity 0 as Note Off, send unmapped notes to the synthesiser, apply aftertouch (0xA0) to the filter, and offer a keyboard fallback and dropped-file samples.	T	PARTIAL	Automated browser tests with a stubbed Web MIDI PASS (T-020, T-021 handoffs). A real browser, a real device and audible output are NOT RUN.
REQ-035	**English language** All project files, code, comments, reports, generated PCB text, fabrication file names and user-visible strings shall be in English.	I	PASS	I: 0 Portuguese-word matches in firmware (`.ino`, `fm_*.h`), gen_*.py, docs/index.html, README.md, hardware/README.md and docs/manual/MANUAL.md. Silkscreen is English (SCALE, OCTAVE, ORIG, QT, legend, 'copper: bottom side'); fabrication file names are English; no broken relative links (`doc.links.*` PASS).

ID	Requirement	Verification	Status	Evidence
REQ-036	**User and assembly manual** The user and assembly manual shall match the current PCB. It shall cover both module revisions, the JP1 selector and data wires, flashing, first start, a valid pre-power continuity checklist and troubleshooting.	I	PASS	I: PASS (systems inspection at a216c1b, T-077). hardware/README.md (assembly) matches the current PCB: JP1 and two wires, R15/R16 220 Ω, JP1 legend, a continuity test that accounts for the 3.3 kΩ resistors, English silkscreen names, valid links (sim/systems/interface_check.out: `SUMMARY documentation: 17 checks, 17 PASS, 0 FAIL`). ADR-021 items: the BOM lists C1 as KEMET C322C104K5R5TA with 5.08 mm leads (and warns that C320 has 2.54 mm leads), assembly step 3 places C1 above RV1; JP1 is described with 3.81 mm pitch and a 1.87 mm gap, bridged with an R13–R16 off-cut (0.5 mm) or solder (BOM row and §5 step 1); the pad rules give J2 oval 3.0 × 4.5 mm. The drill table 0.8/0.9/1.0/1.1/1.4/1.8/3.2 mm = 13/9/63/28/14/2/4, 133 holes in total, equals the Excellon file hardware/fabrication/gerbers/frutamidi.drl counted by systems. C2-001 (R1–R12 lead bend, EE-S27) and C2-007 (JP1 bridge material) stay fixed. README.md documents the breadboard 4.7 kΩ pull-ups. docs/manual/MANUAL.md (user manual) refers to JP1 and covers BOOTSEL/UF2 and troubleshooting (not re-read in T-077). The inspection covered consistency with the design, not readability; the lead forming, the C1 fit and the JP1 bridge are not test-fitted on real parts (physical plan).
REQ-037	**Sensing range (physical)** [PHYSICAL] Touch detection shall work on objects from a bare pad (about 20 pF) to large fruit or a glass of water (up to about 500 pF, ASSUMPTION), with leads of 20–50 cm, after recalibration.	T [PHYSICAL]	NOT RUN	A: the MPR121 can charge up to about 1203 pF (CALCULATION). Latency targets hold up to 210 pF at 400 kHz. AUTOCONFIG limits USL/TL/LSL = 200/180/130 at design VDD 3.2 V and SCTS = 0 (0x7C D7) are now SOURCE (ADR-010/011, MPR121 datasheet S-006 in management/SOURCES.md). T: NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-038	**False triggers and crosstalk (physical)** [PHYSICAL] There shall be no false Note On in 10 min idle with all 12 electrodes connected to objects, and no crosstalk when touching a neighbouring electrode.	T [PHYSICAL]	NOT RUN	T: NOT RUN [PHYSICAL]. Plan addition (C2-012): the CR-002 relayout runs SDA about 3.2 mm from the electrode pads over about 31 mm (reviewer MEASUREMENT on the design file), not analysed for coupling. Test: with all 12 electrodes connected and untouched, log the MPR121 filtered data and baseline of each electrode (a) with normal 400 kHz polling and (b) with I^{2}C traffic reduced to one read per 100 ms (debug build); proposed criterion: the peak-to-peak difference between (a) and (b) is below half RELEASE_ THRESHOLD, i.e. 3 counts (ASSUMPTION; RELEASE_ THRESHOLD 6 and TOUCH_THRESHOLD 12 are SOURCE firmware/ FrutaMIDI/FrutaMIDI.ino), and no false Note On in 10 min. Shares setup with plan item P-4.
REQ-039	**JP1 cannot short 3V3** No position of the JP1 3V3 selector (ORIG, QT, open) shall connect +3V3 to GND.	I	PASS	S1 and S7 are never GND in either revision (`pcb.JP1_no_ gnd_reachable`).
REQ-040	**JP1 wrong-position tolerance** A wrong JP1 position (ORIG on a STEMMA QT module, or QT on an original module), or both sides bridged, shall not apply a voltage outside the MPR121 absolute maximum ratings (inputs $\leq$ VDD + 0.3 V, S-04 Table 3), and shall not short +3V3 through a module output.	A	FAIL	A: still FAIL as stated. The ORIG bridge on a QT module puts +3V3 on ADDR, and the QT bridge on an original module puts it on IRQ, both above VDD + 0.3 V on an unpowered module. Both sides bridged on an original module ties +3V3 to the IRQ open drain. Mitigation per ADR-009 D-10: silkscreen legend present (`pcb.JP1_ legend` PASS), manual warning, and the firmware 'sensor not found' LED pattern. Residual risk R-007 accepted by the manager.
REQ-041	**I2C rise time (breadboard, with pull-ups)** The breadboard-build I2C bus, with the two 4.7 kΩ pull-ups required by ADR-009 D-11, shall meet the rise time for 400 kHz.	A + T [PHYSICAL]	PARTIAL	A: PASS. t_r 156 ns at 60 pF with 4.7 kΩ (CALCULATION). Without them, 452 ns: FAIL above 39.8 pF. I: README.md lists the two 4.7 kΩ pull-ups in its BOM and wiring table. T: NOT RUN (plan item P-4).

ID	Requirement	Verification	Status	Evidence
REQ-042	**Shared-note reference counting** When two or more pads map to the same MIDI note, the note shall be sent once (Note On on the first pad), stay sounding while any of those pads is held, and get its Note Off only when the last of them is released (reference counting). Aftertouch for that note shall be the maximum of the holding pads' values, with a final pressure of 0 on the last release.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS: `tests/firmware/logic/test_notes_health.py` (reviewer sequence, `releaseAll`), `firmware/FrutaMIDI/fm_notes.h` (T-044, 295 passed). Physical NOT RUN.

Status counts (42 requirements): PASS 17, PARTIAL 18, FAIL 3, NOT RUN 4.

3 Architecture and decisions

3.1 System overview

Touch path: electrode object → lead → screw terminal → 3.3 kΩ series resistor (PCB) → MPR121 electrode pin; the MPR121 reports touch bits and filtered/baseline data over I^2C at 400 kHz to the Pico, which sends USB MIDI 1.0 messages to the host (browser sampler or DAW) [`management/INTERFACES.json`] [`reports/systems/architecture_options.md`].

3.2 Architecture options

Three options were compared on the same criteria [`reports/systems/architecture_options.md`]; option A was chosen (ADR-006).

Criterion	A: Pico + MPR121 (chosen)	B: Pico native sensing	C: Pro Micro + MPR121
Core BOM cost	23.60–34.60 EUR (CALCULATION, budgets.out, revision 1)	14.20–22.20 EUR (CALCULATION from ASSUMPTION prices)	22.60–34.60 EUR (ASSUMPTION clone price)
Sensing with fruit/water	Good; hardware autoconfiguration and baseline filter	Unproven; filtering in firmware	Same as A
Latency floor	4 ms at the sensor	Potentially below 1 ms (ASSUMPTION)	Same as A
Constraint compliance	Uses the fixed bought-in parts	Drops the Adafruit 1982 (definition change)	Drops the Pico family (definition change)
Risk level	Low–medium	High	Medium

The BOM figure for A above is the T-008 value; the current core BOM is 23.98–38.62 EUR (CALCULATION, budgets revision 7 after CR-002 and ADR-021, REQ-032, Section 2) [`reports/systems/budgets.md`].

3.3 Decisions

Summary of [`management/DECISIONS.md`] (ADR-000 to ADR-021). Decisions taken by the manager under ADR-001 are flagged there for later user review.

ADR	Decision	Basis / evidence
ADR-000	Project folder, all files in English, existing paths kept, commit rules.	User, Phase A
ADR-001	Run gates G1–G4 end to end; publishing a release still needs explicit user approval.	User
ADR-002	Firmware is the source of truth for the long press: docs state 0.8 s (firmware `> 800` ms).	T-030 handoff
ADR-003	No "safe to touch" claim without evidence; wording follows the electronics analysis.	T-050
ADR-004	English file and folder names (`hardware/fabrication/`, `images/`).	T-051
ADR-005	Sampler: learning a used note swaps the two pads; notes flash before audio is enabled; omni channel; English UI and silkscreen.	T-020
ADR-006	Architecture option A (Pico + Adafruit MPR121 breakout); MPR121 sets a ≥ 4 ms sensing floor.	architecture_options.md
ADR-007	Latency ≤ 15 ms worst case and jitter ≤ 10 ms (relaxed, flagged); burst read M1; 24 pads breadboard only; JP1 3-pad selector replaces the 3V3 wire; hosts; BOM target ≤ 30 EUR; English.	T-008
ADR-008	3.3 kΩ series resistor per electrode (recorded as PASS ±4 kV, later downgraded to an estimate by review F-002; PARTIAL 8 kV, risk R-006); keep 400 kHz with 4.7 kΩ pull-ups; label v1.1 (superseded by ADR-019 D-2).	analysis.md
ADR-009	PARTIAL policy; 220 Ω I^2C series resistors (fault current $\leq$ 15 mA); JP1 silkscreen and warnings (R-007); breadboard 4.7 kΩ pull-ups.	T-010
ADR-010	Run-based burst reads (12 pads 945 µs vs 2430 µs, CALCULATION); AUTOCONFIG limits 200/180/130 at design VDD 3.2 V; SCTS = 0.	T-041
ADR-011	MPR121 register facts confirmed from the datasheet (S-006); three firmware ASSUMPTIONs become SOURCE.	T-042, manager
ADR-012	No datasheet-rated pulse resistor found; MFR-12 3k3 kept, REQ-023 PARTIAL, candidate CR-001 (superseded by CR-002 and ADR-019 D-1); R15/R16 220 Ω accepted.	T-054, T-055
ADR-013	Interfaces frozen as frozen-1; REQ-040 stays FAIL with accepted risk R-007.	Manager
ADR-014	All 16 review findings of 72f67f0 dispositioned (13 FIX, 1 ACCEPTED, 2 info FIX); fixes made on main before the baseline; no frozen interface changed.	Review T-060
ADR-015	BOM part numbers: J2 Phoenix Contact 1715721 (×7); RV1 named by series (RK09K, order code not sourced, PARTIAL accepted); female headers generic; SW1 Omron B3F-1000.	T-056
ADR-016	REQ-018 stays FAIL with a documented deviation (low-power USB port 4.40 V below the Pico's recommended 4.50 V; guide recommends a standard port or powered hub); documentation-only interface change applied as frozen-2.	T-012
ADR-017	Firmware review fixes: per-note reference counting (F-004), sensor health with loss detection and retry (F-005), `setClock` after every sensor `begin()` (F-006).	T-044
ADR-018	Re-review dispositions: F-007 fixed (build logs committed); N-001 accepted as a REQ-016 exception (lost-sensor restart 16.8–22.1 ms, fault path only); N-002 evidence refreshed; N-003 interfaces frozen-3 (IF-PCB-AUX).	T-061

ADR	Decision	Basis / evidence
ADR-019	CR-002 decisions. D-1: ESD scope stays ±4 kV; R1–R12 become Vishay MBE04140C3301FC100 (DIN0414, 3k3 1 %), 4.08 kV 1.2/50 µs rating at 3.3 kΩ (SOURCE EE-S17b), margin 1.12 at 4 kV (CALCULATION); 8 kV stays risk R-006, LSR37 deferred. D-2: board label "FrutaMIDI v1.2". D-3: cost increase accepted, REQ-032 stays FAIL. D-4: relayout accepted (module, J1, JP1, R13–R15, H4 moved; nets and pinout unchanged).	T-060, T-070
ADR-020	CR-002 re-review dispositions: end-cap gap risk (C2-002) controlled by forming the leads on a gauge and a caliper check in the physical plan; the E-pad to GND edge track stays as an intentional spark gap (C2-005); H3 moved to (96, 38) mm.	T-073
ADR-021	Manager PCB design review follow-up: JP1 pitch 3.81 mm (gap 1.87 mm); J2 pads 3.0 × 4.5 mm oval; RV1 tabs on AGND; C1 100 nF wiper filter; silkscreen overlaps removed; C2 1 µF rejected. Addendum: C1 cost accepted (REQ-032 stays FAIL); INTERFACES frozen-6.	T-076, T-077

Change requests. The candidate CR-001 (pulse-rated resistor, not opened) is superseded by CR-002 [`changes/CR-002.md`], which handles the pulse rating together with the flashover gap across the resistor (ADR-019, Section 5.5). The Vishay MBB 0207 named in CR-001 is rated only 1.74 kV at 3.3 kΩ (SOURCE EE-S17b), below the 3.64 kV needed (CALCULATION).

4 Interfaces

Source: [`management/INTERFACES.json`], revision `frozen-6` (CR-002 manager PCB review follow-up, FrutaMIDI v1.2, ADR-021; frozen by the manager on 2026-11-07). History: frozen-1 of 2026-11-03 (ADR-013); documentation-only changes from T-012 (ADR-016) and review N-003 (ADR-018) gave frozen-3 of 2026-11-04. Frozen-4 changes the R1–R12 part and footprint, the IF-TERMINALS protection text, the board revision label, and records the v1.2 placement for traceability; nets, pinout, terminal map, I^2C, USB MIDI and build flags are unchanged. Frozen-5 (T-074) refreshed the evidence notes and the H3 placement after the re-review. Frozen-6 (T-077) adds C1 (100 nF, pot wiper to AGND), puts the RV1 mounting tabs on AGND (AGND is now exactly A1 pin 33, RV1 pin 3, the RV1 tabs and C1 pin 2), and records the JP1 footprint (3.81 mm pitch, 1.87 mm pad gap) and the 3.0 × 4.5 mm oval J2 pads; pinout, terminal map, I^2C, USB MIDI and build flags are unchanged. Changes go through the engineering change process.

Interface	Summary	Requirements
IF-USB-MIDI	Full-speed USB, Audio/MIDIStreaming USB MIDI 1.0, bulk IN/OUT 64-byte packets; bus powered, bMaxPower 100 mA (SOURCE S-11); product string "FrutaMIDI". Sends Note On (0x9n), Note Off (0x8n) and Polyphonic Key Pressure on channel `MIDI_CHANNEL` (default 1, range 1–16); incoming MIDI is read and discarded. Hosts: Windows 10/11, macOS 12 or later.	REQ-011, 012, 013, 019
IF-I2C	400 kHz; sensor 0 at 0x5A, optional sensor 1 at 0x5B (breadboard). PCB pull-ups R13 (SCL), R14 (SDA) 4.7 kΩ to +3V3, plus module 10 kΩ pull-ups and BSS138 shifters. 3.3 V logic. Series R15 (SCL) and R16 (SDA) 220 Ω on the Pico side; fault current 15.5 mA per line (CALCULATION at 3.4 V, ≤ 50 mA IIOVDD, SOURCE S-17).	REQ-021, 025, 026, 031

Interface	Summary	Requirements
IF-PICO-PINS	Breadboard: SDA GP4 (pin 6), SCL GP5 (pin 7), button GP15 (pin 20), pot GP26/ADC0 (pin 31, optional), 3V3 pin 36. PCB: SDA_PICO GP26 (pin 31) via R16, SCL_PICO GP27 (pin 32) via R15, button GP22 (pin 29), pot GP28. AGND (pin 33) is the analogue ground return (SOURCE S-01 §2.1); on the PCB it reaches only RV1 pin 3, the RV1 mounting tabs and C1 pin 2 (ADR-021). LED GP25 (Pico, Pico 2) or CYW43 WL_GPIO0 (W boards) (SOURCE S-02).	REQ-017, 031
IF-TERMINALS	J2: 14 terminals at 5.08 mm: GND, E0–E11, GND; pads 3.0×4.5 mm oval, drill 1.4 mm, copper extended 0.75 mm towards the back of the block (ADR-021). Terminal k ($k = 2..13$) reaches MPR121 ELE($k-2$) through R1–R12 3.3 kΩ, Vishay MBE04140C3301FC100 (DIN0414, lying flat, 15.24 mm lead pitch); protection at ±4 kV is an estimate on ASSUMPTIONs, not a pass (review F-002); resistor pulse rating FAIL at 8 kV (R-006).	REQ-022, 023, 030
IF-MPR121-MODULE	Socket A2: 1×13 electrode row and 1×7 power/I^{2}C row S1–S7. JP1 (upright since v1.2: QT top, ORIG bottom; 3.81 mm pitch, 1.87 mm pad gap since ADR-021): pad 1 S1 (ORIG), pad 2 +3V3, pad 3 S7 (QT); a wrong choice puts +3V3 on ADDR or IRQ, never GND (REQ-039 PASS, REQ-040 FAIL). J1: pad 1 SCL, pad 2 SDA; wires to S5/S6 (original) or S3/S4 (STEMMA QT).	REQ-024, 025
IF-CONTROLS	SW1 to GND with internal pull-up, sampled every 10 ms; long press > 800 ms recalibrates, short press selects the next scale. RV1 10 kΩ linear, wiper filtered by C1 100 nF to AGND ($\tau \leq$ 0.38 ms, CALCULATION): 10-bit ADC read every 30 ms, 5 zones of 0.66 V, hysteresis 0.7 zone, dead band 0.132 V against an error budget of 0.0364 V (CALCULATION, budgets.md).	REQ-006, 007, 008
IF-FW-BUILD	arduino-cli 1.5.1, arduino-pico 6.3.1, Adafruit MPR121 1.2.1, BusIO 1.17.4, MIDI Library 5.0.2. FQBN `rp2040:rp2040:<board>:usbstack=tinyusb` for Pico, Pico W, Pico 2, Pico 2 W; `-DFRUTAMIDI_PCB` selects the PCB pin map; optional `arduino:avr:leonardo`. Artifacts `FrutaMIDI-<board>.uf2` and `FrutaMIDI-pcb-<board>.uf2`.	REQ-033
IF-PCB-AUX	Documentation-only (review N-003): J3 = the S1–S7 wire holes (FrutaMIDI:WireHoles_1x07_P2.54mm), each joined to the module power/I^{2}C row; the female-header sockets (2×1×20 Pico, 1×13 and 1×7 module) are a Socket field of A1/A2, not separate symbols.	—

Board: outline 100 × 70 mm, copper on B.Cu only, label "FrutaMIDI v1.2" (ADR-019 D-2; text in `hardware/scripts/gen_pcb.py` and `hardware/frutamidi.kicad_pcb`), fabrication folder `hardware/fabrication`. The placement recorded in frozen-6 (for example A2 at (60.02, 17.645) mm, H4 at (88.5, 30.5) mm, C1 at (31.2, 5.3) mm, MEASUREMENT on the design file) is for traceability and is not a frozen mechanical interface. Mounting hole H3 moved from (95, 48) to (96, 38) mm in T-073 (review finding C2-009); the placement record in `management/INTERFACES.json` was updated by systems in T-074, and C1 was added in T-077.

5 Electronics analysis

Sources: [`reports/electronics/analysis.md`] (T-050/T-052 and, for CR-002, §5 by T-060 and T-073 and §6 by T-076), handoffs T-053, T-054, T-055, T-060, T-073 and T-076 in [`management/handoffs/electronics-engineer/`], raw outputs in [`sim/electronics/calc/`]. Nothing in this sec-

tion is a measurement on hardware; "MEASUREMENT (design file)" means a dimension read from the KiCad file by a script.

5.1 Schematic and PCB checks

The schematic and PCB are generated by `hardware/scripts/` and checked by `build_all.sh` with kicad-cli 10.0.4. For v1.2 after the manager PCB review follow-up (T-076, committed in a216c1b; systems checks T-077, committed in f457bfe) [`hardware/build/erc.rpt`] [`hardware/build/drc.rpt`] [`reports/electronics/evidence/T-076/`]:

- ERC: "ERC messages: 0 Errors 0 Warnings 0".
- DRC with schematic parity: "Found 0 DRC violations", "Found 0 unconnected pads", "Found 0 Footprint errors".
- Home-etch rules (`sim/systems/pcb_rules_check.py`): "SUMMARY: 19 checks, 19 PASS, 0 FAIL", including "ADR-021 JP1 pad gap >= 1.8 mm" (gaps 1.87/1.87 mm) and "ADR-021 J2 pads 3.0 x 4.5 mm" (14 pads) [`sim/systems/pcb_rules_check.out`].
- Interface check: "SUMMARY: 85 checks, 85 PASS, 0 FAIL", including `pcb.AGND_membership` (AGND pads A1/33, C1/2, RV1/3, RV1/MP) [`sim/systems/interface_check.out`]. The intermediate T-076 run gave 79/80 with the expected FAIL of the old rule `pcb.A1.AGND_isolated`, which T-077 replaced.
- Silkscreen (`check_silk.py`, now also footprint designators and text-to-text): "PASS worst silkscreen-text to pad gap 0.35 mm ('R13' <-> R13 pad 1), 34 texts" and "PASS worst silkscreen text-to-text gap 0.11 mm" (MEASUREMENT, design file) [`reports/electronics/evidence/T-076/check_silk.out`].

These are design-rule results, not hardware validation.

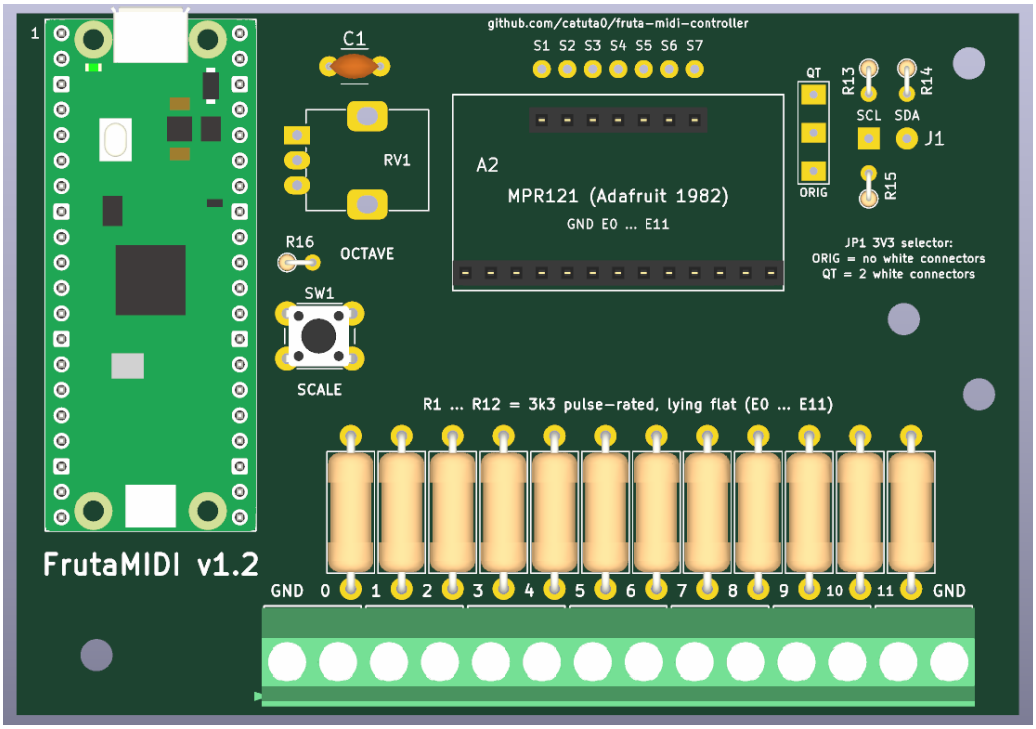


Figure 1: FrutaMIDI v1.2, component side (KiCad 3D render). R1–R12 lie flat in one row above the screw terminals; JP1 stands upright (QT top, ORIG bottom) to the right of the module on a 3.81 mm pitch (ADR-021), with J1 and R13–R15 next to it. C1 (100 nF) sits just above the octave potentiometer RV1. Mounting hole H3 sits at the right edge above R12, at (96, 38) mm since T-073. Render at a216c1b. [`hardware/fabrication/images/board-top.png`]

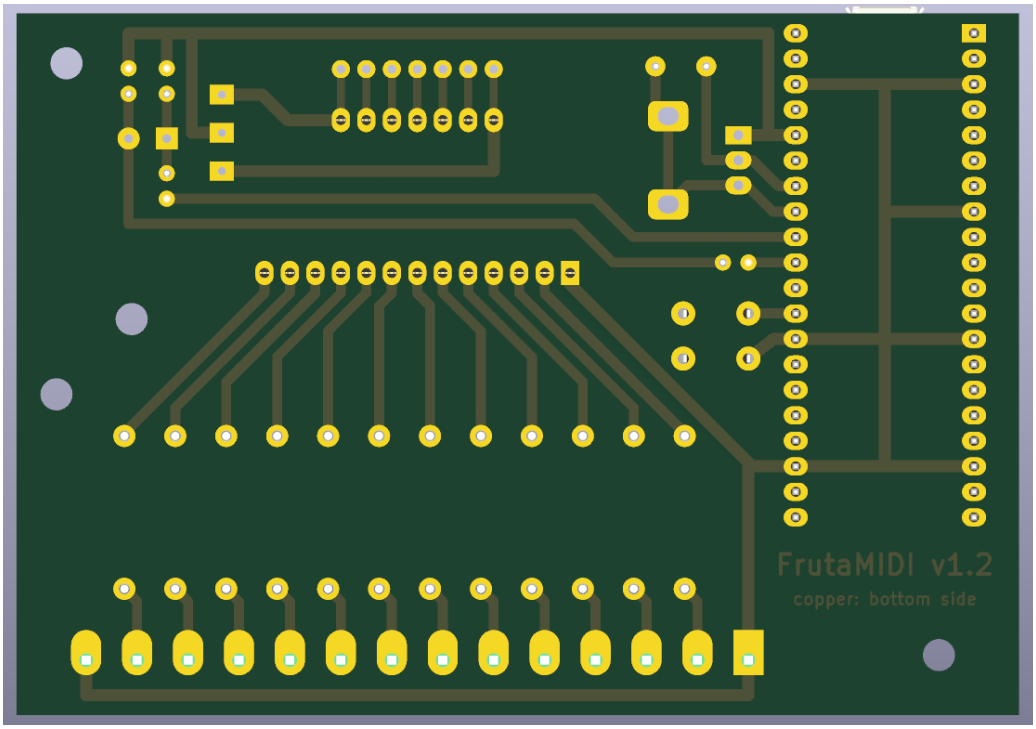


Figure 2: FrutaMIDI v1.2, copper side (bottom view). The ELE fan-out runs at 45° above the resistor row; the E side runs straight down to J2, whose pads are 3.0×4.5 mm ovals extended towards the back of the block (ADR-021). Render at a216c1b. [`hardware/fabrication/images/board-copper.png`]

5.2 Datasheet checks and budgets

- Pico pins (GP26/GP27 I^2C1 on pins 31/32, GP22 pin 29, GP28 ADC2 pin 34) match the firmware PCB map: PASS (SOURCE EE-S08).
- MPR121 VDD through the module regulator: AP2112 ≈3.295 V typical, ≥3.19 V worst; MIC5225 ≈3.25 V typical, ≥3.10 V worst; inside 2.0–3.6 V: PASS (CALCULATION).
- 3V3 load: 4.45 mA worst case, board power 14.7 mW (CALCULATION, electronics). Systems gives 10.15 mA worst case with 2 sensors and PCB pull-ups, 96.6 % margin to 300 mA (CALCULATION) [`reports/systems/budgets.md`] [`management/REQUIREMENTS.md`] (REQ-020).
- VBUS current worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W against 100 mA (CALCULATION; upper bounds ASSUMPTION) [`reports/systems/budgets.md`].

5.3 Bill of materials and cost

Part numbers from [`hardware/fabrication/bom-kicad.csv`] (ADR-015, ADR-019): A1 Raspberry Pi SC0915; A2 Adafruit 1982; **R1–R12 Vishay MBE04140C3301FC100** (3k3 1 %, DIN0414, pulse-rated; was Yageo MFR-12FTF52-3K3 up to v1.1); R13/R14 MFR-12FTF52-4K7; R15/R16 MFR-12FTF52-220R; J2 Phoenix Contact 1715721 (×7); RV1 Alps Alpine RK09K series, 10 kΩ linear, vertical (order code not sourced); **C1 KEMET C322C104K5R5TA** (100 nF X7R 50 V ±10 %, radial, 5.08 mm leads; new in ADR-021; not the C320 version, which has 2.54 mm leads, SOURCE EE-S29); SW1 Omron B3F-1000; female headers generic (a Socket field of A1/A2). JP1 is copper only, and the J2 pad change costs nothing [`management/handoffs/electronics-engineer/T-076.md`].

Item	v1.1 (budgets rev. 5)	v1.2 (budgets rev. 7)	Label
R1–R12, 12 parts	0.24–0.60 EUR	3.60–7.92 EUR	v1.2: SOURCE EE-S26 sibling values, equal 3k3 price ASSUMPTION; v1.1: ASSUMPTION
C1 100 nF (ADR-021)	—	0.20–0.90 EUR	ASSUMPTION from a distributor search, EE-S30, not verified
Core BOM total	20.42–30.40 EUR	23.98–38.62 EUR	CALCULATION
Over the 30 EUR target (upper bound)	0.40 EUR	8.62 EUR	CALCULATION
REQ-032 status	FAIL	FAIL	—

Sources: [`reports/systems/budgets.md`] [`management/REQUIREMENTS.md`] (REQ-032) [`management/metrics.json`] (`bom_core_range`, `pot_filter_C1_price`). The R1–R12 increase is accepted for ESD safety (ADR-019 D-3) and the C1 increase by the ADR-021 addendum; the requirement stays FAIL and is not waived. Revision 6 (before C1) gave 23.78–37.72 EUR. Most other prices are ASSUMPTIONs; a quote from one supplier would narrow the range.

5.4 Electrode path (critical block)

Model `sim_electrode.py` (SIMULATION): constant-current charge into a 17 pF pin plus series R plus external C, swept over R = 0–100 kΩ and C = 10–2000 pF; the discharge model is an ASSUMPTION [`reports/electronics/analysis.md`].

Variant	Peak 4 kV	Peak 8 kV	Sensitivity	Verdict
a) none	15.0 A	30.0 A	1.000	FAIL (11× over)
b) 3.3 kΩ series (selected)	1.10 A	2.20 A	0.904	Estimate at 4 kV (review F-002), PARTIAL 8 kV
b') 10 kΩ series	0.39 A	0.77 A	0.679	FAIL (sensitivity)
c) 3.3 kΩ + TPD4E05U06	1.10 A	2.20 A	≈0.90	FAIL (SMD constraint)
d) 3.3 kΩ + 2×1N4148	1.10 A	2.20 A	≈0.90	PARTIAL (12 jumpers)

- ESD model: IEC 150 pF/330 Ω; pin limit 1.33 A inferred from HBM 2 kV (ASSUMPTION). Minimum R: 2.67 kΩ at 4 kV, 5.67 kΩ at 8 kV. Design rule 2.7–3.3 kΩ; chosen 3.3 kΩ. The v1.2 part keeps the value and the 1 % tolerance, so this sensitivity analysis stays valid.
- With 3.3 kΩ the smallest detectable touch is 0.5 pF on 10 pF and 19.9 pF on 1000 pF (19.8 pF without R) (SIMULATION).
- The breadboard build has no electrode protection beyond the MPR121's internal structures (HBM ±2 kV, SOURCE EE-S01), because the module wires ELE0–11 straight to its header (T-055).

5.5 ESD flashover around the series resistor and pulse rating (CR-002)

Sources: [reports/electronics/analysis.md] §5, [management/handoffs/electronics-engineer/T-060.md], [management/handoffs/electronics-engineer/T-073.md], raw outputs calc_flashover.out, calc_pulse_rated.out, check_gap.out, check_gap_old_86866be.out, check_gap_old_1247c44.out, calc_gnd_bounce.out and calc_mechanical.out in [sim/electronics/calc/]. Change request: [changes/CR-002.md]; decisions ADR-019 and, after the re-review (Section 8.1), ADR-020.

Mechanism. During a contact discharge almost the whole IEC voltage appears across the series resistor, because the pin structure holds the ELE side near 0 V (ASSUMPTION): $V_R = V \cdot R/(R + 330\,\Omega) = 3636$ V at 4 kV and 7273 V at 8 kV (CALCULATION). Two things can then defeat the resistor: an arc across the air gap between its two ends, which leaves the pin with only the 330 Ω source (12.1 A at 4 kV and 24.2 A at 8 kV, against 1.10 A and 2.20 A through R, CALCULATION); or a resistor body not rated for that voltage.

Correction of an earlier claim. The v1.1 analysis stated that "each path still has its series R". That was wrong for the gap across the resistor itself: with the vertical DIN0204 footprint (pads 1.6 mm on 2.54 mm pitch) the ELEx–Ex copper gap was 0.94 mm (MEASUREMENT, design file of 86866be), and the vertical lead ran about 1 mm from the opposite end cap (ASSUMPTION). At an assumed 3 kV/mm breakdown that gap flashes over at about 2.8 kV, below the 3.64 kV across R (CALCULATION, CR-002). The resistor (Yageo MFR-12, 400 V maximum overload, SOURCE EE-S14b) was also not rated for the pulse.

Breakdown sweep and gap rule. Every breakdown model is an ASSUMPTION; the constants are stated in calc_flashover.py. The IEC 60664-1 values were recalled, not re-read (EE-S25), and must be checked against the standard.

Model	Gap at 4 kV (3.64 kV)	Gap at 8 kV (7.27 kV)
Uniform field 4.5 / 3.0 / 2.0 kV/mm	0.81 / 1.21 / 1.82 mm	1.62 / 2.42 / 3.64 mm
Paschen (Townsend form, $\gamma = 0.01$)	0.67 mm	1.56 mm
Empirical uniform air (Kuffel)	0.73 mm	1.79 mm
Non-uniform, $\eta = 0.7$ / 0.5 / 0.3	1.17 / 1.79 / 3.33 mm	2.77 / 4.13 / 7.46 mm
IEC 60664-1 Table F.2 case A, interpolated (recalled)	2.61 mm (3.00 mm at the full 4 kV)	7.07 mm

Design rule (CALCULATION, worst of $\eta = 0.3$ and F.2 case A, rounded up): (a) ELE$x \leftrightarrow$ Ex copper and leads of one channel and (b) E$x \leftrightarrow$ any other channel's ELE net or any MPR121/Pico net $\geq$ **3.5 mm** for 4 kV; 8 kV would need $\geq$ **7.5 mm**. E$x \leftrightarrow$ Ey and E$x \leftrightarrow$ GND are only reported: an arc there still passes a resistor, or diverts the strike to ground.

Part selection. The resistor body must also hold V_R. The 1.2/50 µs ratings at 3.3 kΩ read from the vector chart of Vishay document 28766 (SOURCE EE-S17b; ±5 % reading ASSUMPTION):

Candidate (3.3 kΩ)	Rating	Against 3.64 kV (4 kV case)
Yageo MFR-12 (v1.1 part)	400 V max. overload (EE-S14b)	FAIL
Vishay MBA 0204	1.47 kV	FAIL
Vishay MBB 0207	1.74 kV	FAIL
Vishay MBE 0414, MBE04140C3301FC100 (selected)	**4.08 kV**	PASS, margin **1.12** (1.07 with −5 % reading) (CALCULATION)
Vishay LSR37	12 kV ESD contact (EE-S24)	PASS at 8 kV too, but ±10 % and up to ±20 % drift after 100 strikes

Ratings are SOURCE; the margin is rating/V_R = 4.08/3.64 (CALCULATION).

Pulse energy (corrected in T-073, finding C2-003). The 1.2/50 µs rating pulse puts $(V_{pk}^2/R)\int(V/V_{pk})^2\,dt = 4083^2/3300 \cdot 36.06\,\mu s \approx$ **182 mJ** into 3.3 kΩ (CALCULATION; "Rating pulse energy: integral (V/Vpk)^2 dt = 36.06 us ... -> 182 mJ into 3k3 at 4083 V"). The T-060 value of about 364 mJ used $\int V\,dt$ and was about twice too high. Below its knee the MBE 0414 chart has a log-log slope of 0.496, i.e. constant V^2/R: at 3.3 kΩ the rating is **energy-limited**; above the knee at **5073 Ω** it is voltage-limited at a plateau of **5054 V** (CALCULATION from the SOURCE chart). One 4 kV IEC strike puts 1.09 mJ and 3636 V into R: 167× below the rating-pulse energy and 1.39× below the plateau; at 8 kV, 4.36 mJ and 7273 V exceed both the 3.3 kΩ chart value and the plateau (CALCULATION). The 3.3 kΩ chart value 4083 V stays the acceptance limit (conservative); verdicts unchanged: PASS at 4 kV, FAIL at 8 kV. Taking the 1.2/50 µs rating as a proxy for the ≈0.54 µs IEC pulse is an ASSUMPTION. The MBE 0414 keeps the 3.3 kΩ 1 % value, so the sensitivity analysis above is unchanged (ADR-019 D-1).

Layout (v1.2). R1–R12 lie flat in one row with their axis along y, 15.24 mm lead pitch (datasheet minimum 15.0 mm, SOURCE EE-S17b), pads 2.2 mm, drill 1.1 mm since T-073 (was 1.0 mm; finding C2-008): annular ring (2.2 − 1.1)/2 = 0.55 mm ≥ 0.40 mm and hole − lead = 0.30 mm diametral ≥ 0.2 mm for the 0.8 mm lead (CALCULATION; `calc_mechanical.out` "R_Axial_DIN0414... 1.10 2.20 24 0.55 0.80 0.30 PASS"). The ELE fan-out runs at 45° above the row and the E side runs straight to J2 (Figures 1 and 2). To make room, the MPR121 module moved 15.1 mm up and 2.54 mm right; J1, JP1 (now upright) and R13–R15 moved to the right of the module; H4 moved from (70, 6) to (88.5, 30.5) mm; in T-073 H3 moved from (95, 48) to (96, 38) mm (see "Mounting metal" below). Pinout, nets and the J2 row are unchanged (ADR-019 D-4).

Gaps in the design file. MEASUREMENT on `hardware/frutamidi.kicad_pcb` with `check_gap.py` (KiCad 10.0.4 Python):

Gap	v1.1 (86866be)	v1.2	Against 3.5 mm (4 kV) / 7.5 mm (8 kV)
(a) ELEx ↔ Ex copper, same channel	0.94 mm	13.04 mm (all 12)	v1.1 FAIL; v1.2 PASS / PASS
(b) Ex ↔ other ELE or MPR121/Pico net	1.99 mm	10.92 mm	v1.1 FAIL; v1.2 PASS / PASS
Ex ↔ Ey (reported)	0.80 mm	2.08 mm	reported only
Ex ↔ GND (reported)	1.40 mm	1.40 mm	reported only; intentional spark gap (below)

Proof lines (T-073 re-run, c901ad8): "PASS (a) copper gap 13.04 mm vs rule 7.5 mm (8 kV)", "PASS (b) copper gap 10.92 mm vs rule 7.5 mm (8 kV)"; v1.1 "SUMMARY: FAIL" [`sim/electronics/calc/check_gap.out`] [`sim/electronics/calc/check_gap_old_86866be.out`]. These are copper gaps only. The T-060 "lead ↔ lead 14.44 mm" line was removed in T-073 (finding C2-002): the leads of a flat resistor leave the end caps outwards and are never the closest metal. Copper values are also recorded in [`management/metrics.json`] (`esd_gap_ele_e_min`, `esd_gap_e_other_min`, `electrode_R_pulse_rating_3k3`).

End caps on the component side (T-073, finding C2-002). The closest metal of a flat resistor is cap to cap, not copper to copper. The Vishay drawing gives no cap length, so it is swept: cap length L_c 1.5–3.5 mm (ASSUMPTION), body length 11.9 mm (SOURCE, $L_{\max}$) and 11.0 mm (ASSUMPTION, no $L_{\min}$ given), cap diameter 4.2 mm (SOURCE), lacquer ignored, axial placement error of a hand-fitted body 0 / ±0.25 / ±0.5 mm (ASSUMPTION). Same channel the gap is $L - 2L_c$; the cross-channel pair (E cap of Rk to ELE cap of R$k \pm 1$) is the worst (CALCULATION, `check_gap.out` section (d)):

L / L_c (mm)	Same channel	Cross-channel, no error	±0.25 mm	±0.5 mm
11.9 / 1.5	8.90	8.94	8.45	7.95
11.9 / 2.5	6.90	6.96	6.46	5.97
11.0 / 2.5	6.00	6.06	5.57	5.08
11.0 / 3.0	5.00	5.08	4.59	4.10
11.0 / 3.5 (worst)	4.00	4.10	3.61	**3.13**

Values in mm, CALCULATION on the ASSUMPTIONs above (full sweep in [`sim/electronics/calc/check_gap.out`]). Proof lines: "PASS (d) caps worst 3.61 mm vs rule 3.5 mm (4 kV); FAIL vs 7.5 mm (8 kV) [placement error +-0.25 mm ...]" and "FAIL (d) caps worst 3.13 mm vs rule 3.5 mm (4 kV) ... [placement error +-0.50 mm ...]". Result: at 4 kV **PASS** over the whole sweep for placement error ≤ ±0.25 mm (worst **3.61 mm**); **FAIL** only at the corner L_c 3.5 mm, L 11.0 mm with ±0.5 mm (**3.13 mm**); for $L_c \leq 3.0$ mm it passes at every swept L and error. Status **PARTIAL** (rests on an assumed cap length). Mitigation (ADR-020): the assembly guide asks for the leads to be formed on a gauge so the two straight lengths are equal within about 0.25 mm, and the first-article check measures L and L_c with a caliper (V-26). At 8 kV the 7.5 mm rule is met only by copper; the end caps FAIL ("SUMMARY 8 kV: copper PASS; end caps FAIL (worst 3.13 mm vs 7.5 mm)").

Mounting metal (T-073, finding C2-009). A metal screw head, washer or spacer (radius 2.8–3.5 mm, ASSUMPTION) at a mounting hole can bridge two caps; an arc ELEx → metal → Ex is counted as the sum of the two gaps (ASSUMPTION). At the old H3 (95, 48) mm the bypass sum was 1.94 mm: FAIL against 3.5 mm ([`sim/electronics/calc/check_gap_old_1247c44.out`]). H3 moved to (96, 38) mm, where it faces only the ELE11 cap: worst bypass over H1–H4 "PASS (e) floating-metal bypass worst 15.36 mm vs rule 3.5 mm (4 kV); PASS vs 7.5 mm (8 kV)" (CALCULATION). Nylon hardware at H3 is still recommended in the assembly guide.

Ex ↔ GND 1.40 mm kept as an intentional spark gap (T-073, finding C2-005, ADR-020). Every J2 electrode pad is 1.40 mm (MEASUREMENT, design file) from the GND edge track, below the breakdown distance of every model at 3.64 kV (0.67–3.33 mm), so a 4 kV strike most likely arcs to GND there. That keeps the strike off Rx and the MPR121 pin; moving the track away would put the full 1.10 A into the pin through Rx. The arc current then raises the board ground. GND track lengths (MEASUREMENT, design file): terminal GND to Pico

pin 23 24.04 mm, module GND pin to Pico pin 23 31.36 mm, shared section 4.72 mm. With the IEC 4 kV current (SOURCE EE-S16, recalled from memory: first peak 17.2 A at the upper tolerance, rise 0.6–1.0 ns), track inductance 0.7–1.0 nH/mm and socket plus header pin 3–10 nH (ASSUMPTION), the module-GND to Pico-GND bounce is **87–339 V for ≈1 ns**, about 4.5 V after the first peak; through R15/R16 (220 Ω, then the module's BSS138 level shifters) that is **≤1.53 A** and ≤0.92 nC per I^2C line (CALCULATION on ASSUMPTIONs; "Worst case: module GND - Pico GND bounce 339 V for ~1 ns", [`sim/electronics/calc/calc_gnd_bounce.out`]). The RP2040 HBM 4 kV rating (SOURCE EE-S28) is stated for digital FT pins; no HBM value was found for GP26/GP27, which carry I^2C1 and are ADC-capable (NOT CONFIRMED, escalated). Residual risk: an I^2C pin upset or latch-up from the ≈1 ns bounce. It is an estimate, not a measurement; the ESD test watches for an arc to the GND track and for I^2C errors and resets after each strike (V-10, V-25).

8 kV deferred. At 8 kV the copper gaps meet the 7.5 mm rule but the end caps do not, and the selected resistor does not hold 7.27 kV (4.08 kV rating): FAIL, and the pin current of 2.20 A exceeds the inferred 1.33 A limit by 1.65× (CALCULATION on ASSUMPTION). The scope stays ±4 kV (REQ-023, ADR-019 D-1). The LSR37 (same DIN0414 footprint, 12 kV rated) is a later option; it is deferred because its tolerance and drift bring touch sensitivity to about 0.89–0.90 against the 0.90 criterion (T-060). Risk R-006 (accepted).

Status. Copper flashover around R: PASS on design data at the 4 kV and 8 kV rules. End caps: PARTIAL at 4 kV (assumed cap length and placement), 8 kV not shown. Mounting metal: PASS. Ex ↔ GND: intentional spark gap, residual risk estimated, not measured. Resistor pulse rating: PASS at 4 kV, FAIL at 8 kV. REQ-023 stays PARTIAL until an ESD-gun test (V-10, V-11, V-25, V-26 in Section 10).

5.6 I^2C bus

- At e4668d4 the rise time with only the module 10 kΩ pull-ups was 322 ns at 38 pF against 300 ns: FAIL (CALCULATION, bus C partly ASSUMPTION). With R13/R14 4.7 kΩ: 103 ns at 38 pF (electronics) and 156 ns at 60 pF (systems, RPU 80 kΩ SOURCE S-17): PASS (CALCULATION) [`reports/systems/budgets.md`].
- Series R15/R16 220 Ω: fault current 15.5 mA per line, 30.9 mA for both, against 50 mA (CALCULATION). Low-level margins (authoritative, RP2040 datasheet levels SOURCE S-17): 0.138 V when the Pico drives low (VIL 0.93 V at the MPR121) and 0.235 V when the MPR121 drives low (VIL 0.8 V at the Pico): PASS (CALCULATION) [`reports/systems/budgets.md`]. The earlier electronics figures (0.218 V and 0.286 V, with RP2040 levels as ASSUMPTION, T-055) are superseded interim values.
- Breadboard without the 4.7 kΩ pull-ups: 452 ns at 60 pF, FAIL above 39.8 pF; hence ADR-009 D-11 (CALCULATION).
- v1.2 configuration wires J1 → S holes are about 25–35 mm long (T-060): about 1.8 pF per line at an assumed 50 pF/m (CALCULATION on ASSUMPTION), against the 115.2 pF bus limit at 400 kHz (CALCULATION) [`management/RISKS.md`] (R-008).

5.7 Octave potentiometer filter and AGND (ADR-021)

Sources: [`reports/electronics/analysis.md`] §6, [`management/handoffs/electronics-engineer/T-076.md`], [`sim/electronics/calc/calc_pot_rc.out`], [`sim/systems/budgets.out`] (check A7). C1, 100 nF X7R (KEMET C322C104K5R5TA), runs from the pot wiper (net POT, GP28) to AGND just above RV1; the RV1 mounting tabs are soldered and tied to AGND (symbol `Device:R_Potentiometer_MountingPin`). The AGND net is now exactly A1 pin 33, RV1 pin 3, the two RV1 tabs and C1 pin 2; the only AGND–GND join stays inside the Pico (SOURCE

S-02), and the interface check enforces that no AGND part other than A1 has a GND pad (`pcb.AGND_membership`, `pcb.AGND_no_shared_pad`, `pcb.AGND_parts_not_on_GND`: PASS).

ADC settling. By CALCULATION: the wiper source resistance is $R\,x(1-x)$, largest at mid travel: 2.5 kΩ nominal, 3.0 kΩ worst (10 kΩ +20 %, ASSUMPTION). With C1 at 100 nF nominal and 126 nF worst (+10 % tolerance, +15 % X7R temperature, SOURCE EE-S29):

- $\tau = 0.25$ ms nominal (f_c 637 Hz), **0.38 ms** worst (f_c 419 Hz).
- A full-scale step settles to $\frac{1}{2}$ LSB at 10 bits in $7.62\,\tau = 1.91$ ms nominal, 2.89 ms worst, against the read period `POT_READ_MS` = 30 ms (SOURCE `firmware/FrutaMIDI/fm_pot.h`): margin **10.4**× worst (criterion ≥ 5×, ASSUMPTION).
- Lag during a 100 ms full turn (ASSUMPTION, fastest turn): 3.9 counts worst, against the ≈41-count hysteresis.
- Charge sharing with the RP2040 sample capacitor (≤ 10 pF, ASSUMPTION): ≤ 0.10 LSB.

Status: PASS by calculation ("PASS worst-case settling 2.89 ms vs read period 30 ms"); not measured. REQ-006 stays PARTIAL. The noise benefit of C1 has not been quantified; V-15 and V-28 in Section 10 measure it.

C2 (1 µF at the module) not added. The review proposed a 1 µF decoupling capacitor from +3V3 to GND near JP1. There is no GND copper in the right half of the board above the SDA run; on one layer without vias the only route would pass 3V3 within about 0.05 mm of an assumed H2 washer and run GND about 48 mm under the module, 1.1 mm from all twelve ELE pins. The resulting loop of about 100 mm (≈100 nH at an ASSUMPTIONed 1 nH/mm) would decouple little, and both module revisions carry their own 3.3 V regulator and capacitors (SOURCE EE-S02/EE-S03). Recorded as not needed (ADR-021).

5.8 Safety statement

Each PCB electrode has a 3.3 kΩ pulse-rated series resistor (Vishay MBE04140C3301FC100) lying flat, with at least 13.04 mm between the copper on its two sides (MEASUREMENT, design file); on the component side the end-cap gap is estimated at 3.61 mm or more for a hand-fitted part within ±0.25 mm (CALCULATION on an ASSUMPTIONed cap length, T-073). The J2 electrode pads sit 1.40 mm from the GND edge track, an intentional spark gap that diverts a strike to ground (ADR-020). Its protection against IEC 61000-4-2 contact discharge at ±4 kV is an estimate resting on ASSUMPTIONs (the pin limit is inferred from HBM 2 kV; the breakdown models and the pulse-rating proxy are assumed), not a pass (review finding F-002 [`reports/review/72f67f0/findings.md`]). Only an ESD-gun test can close it. At 8 kV the resistor rating is exceeded (R-006). The design is not evaluated for touching while connected to mains-powered equipment (ADR-003, ADR-012, ADR-019).

6 Firmware design and tests

Sources: [`reports/firmware/coverage.md`] and the T-040 to T-043 handoffs in [`management/handoffs/firmware-engineer/`].

6.1 Design

- One sketch `firmware/FrutaMIDI/FrutaMIDI.ino` with host-testable header-only logic `fm_*.h`: scales (`fm_scales.h`, English names), button (`fm_button.h`), debounce, aftertouch, potentiometer, non-blocking LED blinker (`fm_blink.h`), MPR121 burst reads (`fm_mpr121.h`).

- Pads: Note On at velocity 100 on channel 1 by default; Note Off always carries the stored note; debounce is a 15 ms lockout per pad.
- Aftertouch: Polyphonic Key Pressure 0–127 from delta 12–120; one 0 is sent before Note Off and on a fall to the floor (T-041).
- Scales: 6, Drum Rack (36–47, octave ignored) default; octave $-2..+2$ by potentiometer with 0.7-zone hysteresis.
- Shared notes (T-044, ADR-017): a reference count per note; Note On when the first pad takes a note, Note Off only when the last pad releases it, aftertouch is the maximum of the holders.
- Sensor health (T-044): 3 consecutive failed transactions, or a changed ECR read-back (every 250 ms), mark a sensor lost; its pads are released, the LED shows the fast error blink, and the sensor is restarted after more than 2000 ms (SOURCE `fm_sensor_health.h`). `setClock(400000)` now follows every sensor `begin()` (F-006).
- Defect found (ADR-017): on an I²C NACK the Adafruit library's `touched()` returns 0x0FFF, i.e. all 12 pads touched, so a lost sensor would have sent 12 Note Ons. The sketch now reads the status registers itself and checks the transaction result [`management/handoffs/firmware-engineer/T-044.md`].
- Button: long press > 800 ms recalibrates (releases all notes, restarts sensors, 3 blinks); short press selects the next scale.
- LED patterns are non-blocking: pads stay live while the LED blinks (T-041). `startSensors()` still blocks the loop during a user-requested recalibration (kept by decision; duration to be measured). The automatic restart of a lost sensor also blocks it for 16.8–22.1 ms (CALCULATION), at most once per ≥ 2 s retry; REQ-016 lists this fault-path exception (ADR-018, review N-001).
- MPR121 configuration: AUTOCONFIG with USL/TL/LSL = 200/180/130 at design VDD 3.2 V and SCTS = 0 (ADR-010, ADR-011, SOURCE MPR121 datasheet S-006); I²C stays at 400 kHz; run-based burst reads (worst mask 42 bus bytes in 8 transactions, CALCULATION) [`reports/systems/budgets.md`].

6.2 Latency budget

Budgets revision 5 (T-013, includes the ECR health poll) [`reports/systems/budgets.md`] [`sim/systems/budgets.out`], all CALCULATION for $C \leq 100$ pF at 400 kHz: worst case 10.72 ms (1 sensor) and 12.35 ms (2 sensors) against ≤ 15 ms; jitter 6.72 ms and 8.35 ms against ≤ 10 ms: PASS. A re-touch of the same pad within the 15 ms lockout adds up to 15 ms, giving up to 25.72 ms (1 sensor); REQ-014 is stated for a pad idle for at least 15 ms. At 100 kHz the jitter (1 sensor) and both 2-sensor checks FAIL, so the bus stays at 400 kHz (ADR-008).

6.3 Build and tests

Task	Result line (MEASUREMENT)	Status	Evidence
T-040	Host tests "213 passed in 1.33s"; 14/14 mutants fail the suite; compile matrix 9/9	PASS	`reports/firmware/T-040/`
T-041	Host tests "267 passed in 1.79s"; compile matrix 9/9	PASS	`reports/firmware/T-041/`
T-042	Host tests "268 passed in 1.61s"; datasheet check BLOCKED, later closed by ADR-011	PARTIAL	`reports/firmware/T-042/`
T-043	Host tests "268 passed in 1.97s"; rpipico PCB "Sketch uses 90700 bytes (4%) of program storage space", "Global variables use 17632 bytes (6%)"	PASS	`reports/firmware/T-043/checks.log`

Task	Result line (MEASUREMENT)	Status	Evidence
T-044	Host tests “295 passed in 1.40s”; mutation check 10/10 killed; 9/9 builds exit 0 (rpipico PCB “Sketch uses 91196 bytes (4%)”)	PASS	`reports/firmware/T-044/`

A clean compile and host tests are not hardware validation. The host loop harness copies the call order of `loop()` and does not run the sketch itself; the timing granularity on target depends on the loop period, which has not been measured [`reports/firmware/coverage.md`].

7 Host software and tests

Source: `docs/index.html` (browser sampler, GitHub Pages), handoffs T-020 and T-021 in [`management/handoffs/software-engineer/`], raw outputs in [`reports/software/`].

7.1 Design

- Web MIDI input, omni channel; 12 pads mapped to notes 36–47 by default; Note On with velocity 0 is treated as Note Off (REQ-034).
- Learning a note already used by another pad swaps the two pads' notes; notes received before audio is enabled flash the pad and emphasise the enable-sound button (ADR-005). All UI strings are in English (`lang="en"`).

7.2 Tests

- T-020: `run_pytest tests/host` with Python 3.9.6, pytest 8.4.2, node v26.4.0, playwright-core 1.64.1, Google Chrome 156.0.8078.74: “37 passed in 69.20s” (MEASUREMENT) [`reports/software/T-020/pytest_tests_host_verbose.txt`].
- T-021: same tools, Chrome headless: “102 passed in 56.45s” (MEASUREMENT), including the ADR-005 behaviour tests, a rendered language test over 8 states and mutation checks [`reports/software/T-021/pytest_tests_host_verbose.txt`].

The tests use a stubbed Web MIDI API in headless Chrome. No real MIDI device, no audible output and no person looking at the page in a real browser have been tested (T-020, T-021).

8 Independent review findings

An independent reviewer, who did not produce the work, reviewed revision `72f67f0` (task T-060) [`reports/review/72f67f0/findings.md`] and then re-reviewed the fixes in `72f67f0..05a06a3` (task T-061) [`reports/review/05a06a3/findings.md`]. Dispositions are recorded in ADR-014 and ADR-018 [`management/DECISIONS.md`].

Summary.

- First review: 16 findings (0 critical, 3 major, 9 minor, 4 info); dispositions 13 FIX, 1 ACCEPTED, 2 info FIX (ADR-014).
- Re-review: 14 CLOSED, 1 ACCEPTED (F-012), 1 NOT CLOSED at 05a06a3 (F-007, fixed afterwards, ADR-018); 5 new findings (2 minor, 3 info).
- Reviewer re-checks at 05a06a3: `run_pytest tests/firmware` “295 passed in 1.93s”; ERC “Found 0 violations”; DRC 0/0/0; status counts 17/18/3/4 = 42 match (MEASUREMENT).

Highlight: defect found while fixing F-005. On an I^2C NACK the Adafruit library's `touched()` returned 0x0FFF, i.e. all 12 pads touched, so a lost sensor would have sent 12 Note Ons. The firmware now reads the status registers itself and checks the transaction result (ADR-017). The re-review confirmed the fix (F-005 CLOSED).

Table 12: Review findings, dispositions and re-review verdicts.

ID	Sev.	Finding (one line)	Disposition	Re-review
F-001	major	A wrong JP1 position was called harmless; it exceeds the MPR121 VDD + 0.3 V absolute maximum and back-feeds the module.	FIX: README §5/§6	CLOSED
F-002	major	The ±4 kV "PASS" compared peak current only; pin limit is an ASSUMPTION; resistor not pulse-rated.	FIX: REQ-023 is an estimate, not a pass	CLOSED
F-010	major	REQ-018 was PARTIAL although budget check A3b FAILs (4.40 V < 4.50 V).	FIX: REQ-018 FAIL	CLOSED
F-003	minor	Pot hysteresis defined two ways; offset share misstated.	FIX: REQ-006 restated	CLOSED
F-004	minor	Two held pads can share a note; releasing one sent Note Off.	FIX: reference counting	CLOSED
F-005	minor	Sensor lost during play not detected; missing second sensor never retried.	FIX: sensor health	CLOSED
F-006	minor	Leonardo could run I^2C at 100 kHz after `begin()`.	FIX: `setClock` after every `begin()`	CLOSED
F-007	minor	REQ-033 build logs not in the commit.	FIX: commit logs	NOT CLOSED at 05a06a3; fixed after (ADR-018)
F-008	minor	budgets.md stale; coverage.md "USL 201".	FIX: regenerate, reconcile	CLOSED (residue N-002)
F-009	minor	README named an "aprender" button.	FIX	CLOSED
F-011	minor	Latency model omitted the 15 ms pad lockout.	FIX: REQ-014 scoped to first touch	CLOSED
F-014	minor	Terminal pin diagonal 1.27 mm in a 1.3 mm hole; pot slot drilling unclear.	FIX: 1.4 mm drill, slot instruction	CLOSED
F-012	info	Latency rests on an assumed MPR121 status-update model.	ACCEPTED (plan P-2)	ACCEPTED
F-013	info	ERC ignored single-pin labels globally.	FIX at root (J3)	CLOSED
F-015	info	BOM lacked part numbers; strip count wrong.	FIX	CLOSED
F-016	info	AGND relied on the Pico's internal tie without citation.	FIX: cite S-01/S-02	CLOSED
N-001	minor	Automatic restart of a self-reset sensor blocks the loop 16.8–22.1 ms, beyond REQ-016's 10 ms.	ACCEPTED as specified exception (ADR-018)	new
N-002	minor	REQ-016 evidence stale (worst loop ≈1.72 ms incl. ECR poll).	FIXED (ADR-018)	new
N-003	info	INTERFACES had no J3/J4–J7.	FIXED (frozen-3, IF-PCB-AUX)	new
N-004	info	With 2 sensors configured and one absent the LED blinks fast permanently.	ACCEPTED (matches REQ-009)	new
N-005	info	A second pad on an already sounding note is silent (no retrigger).	ACCEPTED (matches REQ-042)	new

What the review did not cover. First review: INTERFACES.json field by field; visual check of the copper PDFs and mirroring; the sampler beyond its MIDI handler; 8 of 9 firmware targets; MPR121 power-on-reset timing. Re-review: compile runs (logs only); copper PDFs and silkscreen visually; this report's LaTeX; the individual test cases of `test_notes_health.py`. Neither review is a hardware test.

8.1 CR-002 independent re-review

An independent reviewer re-reviewed the CR-002 outputs at `8394b41` on branch `change/CR-002` (task T-071): R1–R12 part and footprint, relayout, gap check, pulse-rating calculation, budgets revision 6, REQUIREMENTS draft-6, INTERFACES frozen-4 and the documentation. The harness refused the reviewer's write of the findings file; the manager recorded the findings from the reviewer's hand-back report without changing their substance [`reports/review/CR-002/findings.md`]. The electronics fixes were made in T-073 (committed in c901ad8) [`management/handoffs/electronics-engineer/T-073.md`]; dispositions are recorded in ADR-020 [`management/DECISIONS.md`].

Summary.

- 13 findings: 0 critical, 0 major, 6 minor, 7 info.
- Dispositions: 8 fixed (C2-006 by systems in T-074, committed in 2ba289a), 1 partly fixed with a residual accepted risk (C2-002), 1 accepted risk (C2-005), 2 accepted (C2-012 with a test added, C2-013), 1 no action (C2-004).
- Reviewer re-checks at 8394b41, all PASS (recorded in the findings file; raw outputs in `reports/review/CR-002/`): ERC 0; DRC 0, unconnected 0, parity 0; `check_gap.py` PASS (copper only); `calc_flashover` and `calc_pulse_rated` outputs byte-identical; interface check 80/80 (committed output stale, 79/80); `pcb_rules_check` 16/16; budgets byte-identical (23.78–37.72 EUR); independent chart readings at 3.3 kΩ 4086 / 1740 / 1469 V for 0414 / 0207 / 0204; drill count 131 equal to the assembly guide; configuration wires 21.1–32.1 mm (MEASUREMENT, design file).
- Reviewer's requirement cross-check: REQ-023 PARTIAL (agreed); REQ-027 PASS; REQ-028/029 PASS with stale evidence text (C2-006); REQ-032 FAIL; REQ-036 should be PARTIAL until C2-001 and C2-007 are fixed (both fixed in T-073).

Highlights. The closest metal of a flat resistor is its end caps, not the copper (C2-002): the 8 kV gap claims now apply to copper only, and the 4 kV end-cap result is PARTIAL on an assumed cap length (Section 5.5). The pulse energy had been overstated about twice (C2-003): 182 mJ, not 364 mJ, with the verdicts unchanged. A metal screw at the old H3 position could bypass R12 (1.94 mm bypass sum, FAIL); H3 moved to (96, 38) mm (15.36 mm, PASS) (CALCULATION, `check_gap.out`).

Table 13: CR-002 re-review findings and dispositions (ADR-020).

ID	Sev.	Finding (one line)	Disposition
C2-001	minor	Assembly guide asked for a lead bend ≥1.5 mm from the body; at 15.24 mm pitch the lead centre is 1.67 mm from the body end (CALCULATION).	Fixed: reworded (Vishay $M_{\min}$ 15.0 mm ≤ 15.24 mm pitch; hold the lead next to the body, test-fit one part).
C2-002	minor	Closest metal of a flat resistor is cap to cap, not the 13 mm copper gap; `check_gap.py` measures copper only; 8 kV gap claims overreach.	Partly fixed, residual accepted risk: cap-to-cap check added; 4 kV PASS for placement error ≤ ±0.25 mm (worst 3.61 mm), FAIL corner 3.13 mm at ±0.5 mm; 8 kV caps FAIL; leads formed on a gauge, caliper check in the plan; 8 kV statements limited to copper.
C2-003	minor	Pulse energy overstated about 2× ($\int V\,dt$ form); rating energy-limited at 3.3 kΩ.	Fixed: 182 mJ ($\int V^2\,dt$); energy-limited below the 5073 Ω knee, plateau 5054 V; verdicts unchanged.
C2-004	info	Independent chart reading confirms 4086 / 1740 / 1469 V; dimensions, LSR37 data and part number agree.	No action.

ID	Sev.	Finding (one line)	Disposition
C2-005	minor	Every E terminal pad is 1.40 mm from the GND edge track, so a strike most likely arcs to GND; the current through the GND trunk to Pico, module and USB was not analysed.	Accepted risk: intentional spark gap; ground bounce 87–339 V for ≈1 ns, ≤1.53 A and ≤0.92 nC per 220 Ω I²C resistor (CALCULATION on ASSUMPTIONed inductances); arc to GND added to the ESD test; GP26/GP27 HBM rating not confirmed (open).
C2-006	minor	Stale evidence: committed `interface_check.out` 79/80; REQ-028/029, requirements.csv and INTERFACES frozen-4 notes describe 3a3684b.	Fixed by systems (T-074): `sim/systems/*.out`, REQUIREMENTS, INTERFACES frozen-5 (committed in 2ba289a).
C2-007	minor	Assembly guide offered any resistor lead off-cut for JP1 (0.8 mm holes); R1–R12 leads are 0.8 mm.	Fixed: solder or an off-cut of an R13–R16 lead (0.5 mm) only.
C2-008	info	1.0 mm drill for a 0.8 mm lead sits exactly on the 0.2 mm fit threshold; generator comment wrong.	Fixed: R1–R12 drill 1.1 mm (ring 0.55 mm); comment corrected.
C2-009	info	Courtyard gaps 0.06 mm (R row to J2) and 0.09 mm (R12 to H3); M3 head about 0.5–0.9 mm from R12.	Fixed: H3 moved (95, 48) → (96, 38) mm; bypass 1.94 mm FAIL → 15.36 mm PASS; nylon hardware recommended.
C2-010	info	Terminal labels "10" and "11" touch R10/R11 pad 2 (about 0.1 mm).	Fixed: worst silkscreen-to-pad gap 0.43 mm.
C2-011	info	`check_gap.py` worst-pair name not deterministic on ties.	Fixed: tie-break by reference, then pad.
C2-012	info	I²C relayout under the module (SDA 3.2 mm from electrode pads over about 31 mm) not analysed.	Accepted, test added: I²C traffic against baseline noise (V-27).
C2-013	info	IEC 60664-1 F.2 values recalled; the 3.5 mm rule comes from model M4 ($\eta = 0.3$).	Accepted: F.2 kept as a cross-check marked "recalled" (EE-S25).

Re-checks after the fixes (T-073, c901ad8). ERC "ERC messages: 0 Errors 0 Warnings 0"; DRC "Found 0 DRC violations", "Found 0 unconnected pads"; "SUMMARY: 16 checks, 16 PASS, 0 FAIL" (home-etch rules); "SUMMARY: 80 checks, 80 PASS, 0 FAIL" (interfaces) [`reports/electronics/evidence/T-073/`]; `check_gap.out` "SUMMARY: copper PASS (4 kV rule, (a) and (b)); end caps PARTIAL at 4 kV over the swept ASSUMPTIONS; mounting metal PASS". These fixes were not re-reviewed independently.

What the re-review did not cover. The findings file does not list exclusions. It is a design-file and document review: nothing in it is a hardware measurement, and it did not see the T-073 fixes or the systems update T-074.

8.2 Manager PCB design review (analisar-pcb) of v1.2

After the CR-002 re-review fixes, the manager reviewed the v1.2 board with the `analisar-pcb` PCB design review procedure. The review is not independent: the manager also directs the design. The electronics engineer acted on its five items in T-076 (committed in a216c1b); systems updated the checks, budgets, interfaces (frozen-6) and requirements (draft-8) in T-077 (committed in f457bfe). Dispositions are recorded in ADR-021 and its addendum [`management/DECISIONS.md`] [`management/handoffs/electronics-engineer/T-076.md`] [`management/handoffs/systems-engineer/T-077.md`] [`reports/electronics/analysis.md`] (§6).

Table 14: Manager PCB design review of v1.2: items and dispositions (ADR-021).

#	Item	Change and evidence	Disposition
1	JP1 pads 0.60 mm apart on a 2.54 mm pitch: one solder blob can join all three pads (3V3 on both S1 and S7).	Pitch 3.81 mm, pads 1.94×2.4 mm, drill 0.8 mm: pad gap 0.60 → **1.87 mm** on both sides, pads 1 to 3 5.68 mm apart (MEASUREMENT, design file); check "ADR-021 JP1 pad gap >= 1.8 mm" PASS.	Fixed. Bridge with solder or an R13–R16 lead off-cut.
2	J2 screw-terminal pads round 3.0 mm: little copper to resist peeling under screw torque on a home-etched board.	Pads 3.0×4.5 mm oval (pad 1 rectangular), drill 1.4 mm, copper offset 0.75 mm towards the back of the block: copper **7.1 → 11.6 mm²** per pad (CALCULATION). Offset rather than centred so that the Ex to GND edge-track spark gap stays 1.40 mm (a centred oval would give 0.65 mm); adjacent pads 2.08 mm, ring 0.80 mm (MEASUREMENT); `check_gap.out` identical to T-073.	Fixed. Peel strength not measured (V-28).
3	RV1 metal tabs unconnected; wiper unfiltered.	Tabs on AGND; C1 100 nF X7R 50 V, KEMET **C322C104K5R5TA** (5.08 mm leads; the C320 has 2.54 mm leads, SOURCE EE-S29), wiper to AGND. $\tau \leq$ **0.38 ms** worst against the 30 ms read period: settling 2.89 ms, margin **10.4**× (CALCULATION, Section 5.7). Interface check rule changed to AGND membership: 85/85 PASS.	Fixed. Cost +0.20–0.90 EUR (ASSUMPTION) accepted; REQ-032 stays FAIL.
4	C2 1 µF from +3V3 to GND at the MPR121 module.	Not added: no single-layer route without running 3V3 next to the H2 fixing metal and a GND track about 48 mm long 1.1 mm from all twelve ELE pins; the ≈100 mm loop would decouple little; both module revisions have their own regulator and capacitors (SOURCE EE-S02/EE-S03).	Rejected, recorded as not needed.
5	Silkscreen overlaps: R16 and SW1 designators overlapped; R14 designator 0.16 mm from R13 pad 1.	Designators moved; JP1 legend line pitch 1.4 → 1.55 mm; `check_silk.py` now also checks footprint designators and text-to-text: worst text-to-pad 0.35 mm, text-to-text 0.11 mm, PASS (MEASUREMENT, design file).	Fixed.

Re-checks after the changes (a216c1b, f457bfe). ERC "ERC messages: 0 Errors 0 Warnings 0"; DRC "Found 0 DRC violations", "Found 0 unconnected pads", "Found 0 Footprint errors" [`reports/electronics/evidence/T-076/`]; "SUMMARY: 19 checks, 19 PASS, 0 FAIL" (home-etch rules); "SUMMARY: 85 checks, 85 PASS, 0 FAIL" (interfaces); budgets "SUMMARY: 30 checks, 25 PASS, 5 FAIL" (the five FAILs are the pre-existing A3b VBUS row, three 100 kHz latency rows and the BOM cost) [`sim/systems/`]. A mutation test on a scratch copy of the board (an SW1 GND pad moved to AGND, JP1 pad 2 shifted) made the new AGND and JP1 checks FAIL as intended (T-077). These changes were not re-reviewed independently, and none of them has been checked on hardware.

9 Limitations

No hardware tests. Nothing has been built, flashed, powered or measured. Building and measuring hardware is out of scope for this project (risk R-004, "Certain", "High") [`management/RISKS.md`]. Every [PHYSICAL] requirement is therefore PARTIAL or NOT RUN, and no statement in this report means that the hardware works.

Analysis-only results. The following results rest only on calculation, simulation, design-rule checks or host tests: latency and jitter (REQ-014/015); VBUS range and current (REQ-

018/019); I²C rise time and logic levels (REQ-026/041, REQ-024); electrode ESD (REQ-023); sensitivity with the series resistors (SIMULATION); ERC/DRC/parity and home-etch rules; firmware behaviour (host tests on a copy of the loop order); sampler behaviour (headless Chrome with a stubbed Web MIDI API).

Open FAILs and accepted risks. REQ-018 (4.40 V low-power-port VBUS minimum below the Pico's recommended 4.50 V; documented deviation, ADR-016); REQ-032 (BOM upper bound 38.62 EUR > 30 EUR after CR-002 and ADR-021, prices mostly ASSUMPTION; increases accepted for ESD safety, ADR-019 D-3, and for the C1 wiper filter, ADR-021 addendum); REQ-040 (wrong JP1 position exceeds MPR121 input ratings on an unpowered module; accepted risk R-007); R-006 (at 8 kV contact the 7.27 kV across each resistor exceeds its 4.08 kV rating, and the pin current exceeds the inferred limit by 1.65×; accepted, ADR-019 D-1). CR-002 re-review (ADR-020): end-cap gap at 4 kV PARTIAL, resting on an assumed cap length (C2-002); Ex ↔ GND spark gap with an estimated 87–339 V ground bounce for ≈1 ns and no confirmed HBM rating for GP26/GP27 (C2-005).

Main assumptions (each to be retired by the plan in Section 10):

- Component prices (BOM, REQ-032) [`reports/systems/budgets.md`].
- Resistor end-cap length 1.5–3.5 mm, body length ≥ 11.0 mm, placement error ≤ ±0.5 mm; track inductance 0.7–1.0 nH/mm and socket plus pin 3–10 nH [`management/handoffs/electronics-engineer/T-073.md`].
- Electrode capacitance: objects up to about 500 pF (REQ-037); 1 m lead 30–50 pF; hand 100–300 pF [`reports/electronics/analysis.md`].
- ESD: ±4 kV target following IEC 61000-4-2 level 2 (the standard was not read); MPR121 pin limit 1.33 A inferred from HBM 2 kV; electrode discharge model; the breakdown models of the gap rule and the IEC 60664-1 Table F.2 values (recalled, not re-read, EE-S25); the resistor chart reading (±5 %); a 1.2/50 µs rating taken as conservative for the ≈0.54 µs pulse; the 3k3 MBE 0414 price equal to its sibling values (EE-S26).
- Board currents (upper bounds), Schottky diode drop and regulator efficiency [`reports/systems/budgets.md`].
- I²C bus capacitance; Pico pin capacitance; in the electronics calculation, RP2040 VIL 0.8 V and VOL 0.4 V and BSS138 $R_{DS(on)} \leq 10\,\Omega$ (T-055).
- MIC5225 typical dropout from a search extract only [`reports/electronics/analysis.md`].
- Firmware timing: loop period "well under 15 ms", button bounce < 10 ms, header and pot pin sizes for the 1.0 mm holes [`reports/firmware/coverage.md`].

Process limitations. Requirements are draft-8 (T-077), not baselined. The independent re-review of the CR-002 changes is reported in Section 8.1; the later manager PCB review and its fixes (ADR-021, Section 8.2) were not re-reviewed independently. The committed systems outputs are current at f457bfe (interface check 85/85, home-etch rules 19/19, budgets 25 PASS and 5 FAIL of 30). Some user-documentation statements (prices, OS behaviour, etching process values) are not traced to a source (documentation handoffs T-030 to T-035).

9.1 Risk register

From [`management/RISKS.md`]. CR-002 rewrote R-006 and added R-008 to R-010; ADR-021 added no risk.

ID	Risk	L / I	Mitigation	Status
R-001	Two Adafruit 1982 revisions with different power-row pin order; wrong wiring shorts 3V3 to GND	M / H	Configuration wires and inspection table	OPEN
R-002	Large conductive objects may exceed the MPR121 range or be unstable	M / M	AUTOCONFIG; threshold sweep; validation plan	OPEN
R-003	Home-etch defects (bridges between 2.54 mm pads)	M / M	0.6 mm clearance rule; continuity checklist	OPEN
R-004	Nothing tested on real hardware (out of scope)	Certain / H	Physical validation plan; no readiness claims	OPEN
R-005	Limited disk space on the host	M / M	Install only what is needed	OPEN
R-006	ESD at 8 kV: 7.27 kV across each 3k3 (CALCULATION) exceeds the 4.08 kV rating of the MBE 0414 (SOURCE EE-S17b); pin current 2.20 A vs inferred 1.33 A (1.65×). Copper gaps meet the 8 kV 7.5 mm rule; the end caps do not (T-073, finding C2-002)	L / M	Scope stays ±4 kV (ADR-008, ADR-019 D-1); later option Vishay LSR37 (same footprint, 12 kV), deferred for tolerance and drift; IEC test in the plan	ACCEPTED
R-007	Wrong JP1 position puts +3V3 on ADDR (QT) or IRQ (original) of an unpowered module	L / L–M	Silkscreen, manual warning, LED error pattern (ADR-009)	ACCEPTED
R-008	v1.2 relayout lengthens the two J1 → S configuration wires to about 25–35 mm: a loose wire can be pushed onto the wrong S hole or strain its joint. About 1.8 pF per line (CALCULATION, 50 pF/m ASSUMPTION) vs the 115.2 pF limit; a wrong hole is limited to 15.5 mA by R15/R16	L / L	Wiring table, pre-power inspection; documentation says to route the wires flat and short; physical check in the plan	OPEN
R-009	End-cap flashover around R1–R12 at 4 kV depends on lead forming and on the unsourced end-cap length (C2-002): cap-to-cap worst 3.61 mm at ±0.25 mm placement error (PASS against 3.5 mm), 3.13 mm at ±0.5 mm (FAIL) (CALCULATION on ASSUMPTIONs)	M / M	Leads formed on a gauge; caliper check of one MBE 0414 before the ESD test (V-26); REQ-023 stays PARTIAL (ADR-020)	ACCEPTED
R-010	Electrode pads 1.40 mm from the GND edge track (intentional spark gap, C2-005): estimated 87–339 V ground bounce for ≈1 ns, ≤1.53 A and ≤0.92 nC per I^2C line (CALCULATION on ASSUMPTIONs); no HBM rating found for GP26/GP27	L / M	Spark gap kept (ADR-020; also kept by the J2 pad change of ADR-021); firmware recovers a lost sensor; ESD test watches for arcs, I^2C errors and resets (V-25)	OPEN

10 Physical validation plan

No part of the hardware has been measured. Each row is a physical test still to run. Pass criteria are the requirement thresholds in [`management/REQUIREMENTS.md`]; where a requirement gives no threshold, the analysis prediction is the criterion and is labelled. Items P-1 to P-6 are the systems plan [`management/handoffs/systems-engineer/T-011.md`]; hardware items from firmware are in [`reports/firmware/coverage.md`]; electronics next tests are in the T-050 to T-055, T-073 and T-076 handoffs. All rows are NOT RUN.

Table 16: Physical validation plan (all NOT RUN).

ID	What to measure	How / instrument	Pass criterion	REQ
V-01 (P-1)	Object plus lead capacitance	LCR meter on typical objects (bare pad, fruit, glass of water) with 20–50 cm leads	Touch detected after recalibration from about 20 pF to about 500 pF (ASSUMPTION range in REQ-037)	REQ-037
V-02 (P-2)	Touch-to-USB latency and jitter	Logic analyser on a test electrode, SDA/SCL and USB	Worst case $\leq$ 15 ms; jitter $\leq$ 10 ms, for $C \leq 100$ pF	REQ-014, 015
V-03 (P-3)	VBUS current per Pico variant	USB inline current meter; Pico, Pico W, Pico 2, Pico 2 W	$\leq$ 100 mA	REQ-019
V-04	VBUS range	Adjustable bench supply at 4.40 V and 5.50 V at the connector	Device enumerates and plays at both ends	REQ-018
V-05 (P-4)	I^2C rise time and low levels, PCB and breadboard	Oscilloscope on SDA/SCL at the Pico and at the module	Rise $\leq$ 300 ns at 400 kHz; low levels below VIL (0.93 V at the MPR121, 0.8 V at the Pico)	REQ-026, 041
V-06	Misplaced data wire current	Wire J1 SDA/SCL to a supply hole; current meter in series	$\leq$ 15.5 mA per line (CALCULATION); within RP2040 IO limits	REQ-024
V-07 (P-5)	Host enumeration	Windows 10, Windows 11, macOS 12 or later; MIDI monitor	No vendor driver; name "FrutaMIDI"; Note On/Off and poly aftertouch on channel 1 only	REQ-011, 012, 013
V-08 (P-6)	False triggers and crosstalk	10 min idle with all 12 electrodes on objects; then touch each pad and watch neighbours	No false Note On; no crosstalk	REQ-038
V-09	Electrode readings with and without R	`SERIAL_DEBUG` deltas per object, with the 3.3 kΩ resistor and with it bridged	Touch detected on every object; delta ratio $\geq$ 0.904 (SIMULATION prediction)	REQ-037
V-10	ESD robustness	IEC 61000-4-2 contact discharge ±4 kV on each terminal, at a test lab	No permanent damage: MPR121 still responds, all pads still detect touch, baseline drift $<$ 1 count (T-060)	REQ-023
V-11	Series resistor after ESD	Watch for flashover across R1–R12 during V-10; measure R1–R12 afterwards	No flashover; each resistance within 1 % of its pre-test value (T-060)	REQ-023
V-12	JP1 bridge and etch gaps	Visual inspection with a magnifier and continuity after etching and bridging	Bridge only on the chosen side; the 1.87 mm gap on the other side clear (ADR-021) and 0.6 mm track gaps clear; no short to GND in any position	REQ-025, 039
V-13	Real loop period	Toggle a spare GPIO each loop; oscilloscope	No blocking $>$ 10 ms during play, except user recalibration and the lost-sensor restart (ADR-018)	REQ-016
V-14	Recalibration duration	GPIO toggle around `recalibrate()`	Recorded per sensor; compare with 16.8–22.1 ms (CALCULATION); no stuck notes afterwards	REQ-008, 002
V-15	Pot ADC noise on the PCB	Log `analogRead` at each zone border for 10 s	Peak-to-peak noise $<$ 41 counts	REQ-006
V-16	Blink with live pads	Touch pads while the LED blinks scale 6; press again during a blink	Notes sent during the blink; pattern restarts with the new scale	REQ-007, 016
V-17	Button bounce and long press	Oscilloscope on the button pin; press tests	Bounce $<$ 10 ms (ASSUMPTION); long press acts after $>$ 800 ms	REQ-007, 008

ID	What to measure	How / instrument	Pass criterion	REQ
V-18	Sensor absent or lost, and LED	Power without the module; then with it; unplug the module during play; every Pico board	No MIDI from the missing sensor, its notes released, LED at 5 Hz, retry every 2 s; 2 slow blinks when found; LED lit while touched (W boards via CYW43)	REQ-009, 010
V-19	MPR121 configuration and burst reads	Read back 0x7B–0x7F after start-up; logic analyser during play	Values as configured; 2 transactions per sensor per aftertouch pass; burst data equal to per-electrode reads	REQ-004, 037
V-20	Sampler in real browsers	Chrome and Edge on Windows and macOS with the device	Pads 36–47 sound; velocity 0 acts as Note Off; learn swaps notes	REQ-034
V-21	Sampler with a busy device	Open the device in a DAW on Windows first, then the sampler	Sampler reports the device as busy	REQ-034
V-22	Shared notes	Set an octave so two pads share a note; touch both, release one, then the other	One Note On; Note Off only after the last release; aftertouch = the stronger pad	REQ-042
V-23	v1.2 configuration wires (R-008)	Inspect the two J1 → S wires before power; continuity J1 SDA/SCL to the S hole of the module revision	Each wire on the correct S hole (S5/S6 original, S3/S4 STEMMA QT), flat and mechanically secure	REQ-024, 025
V-24	R1–R12 fitted as designed	Inspect the resistor row after soldering; caliper on one resistor	Lying flat, part MBE04140C3301FC100, leads 15.24 mm apart, no lead bent towards a neighbour; leads formed on a gauge, the two straight lengths equal within about 0.25 mm (T-073)	REQ-023
V-25	Arc path and I^2C upsets during ESD (C2-005)	During V-10, watch each terminal for an arc to the GND edge track; after every strike check for I^2C NACKs (logic analyser on SDA/SCL), the lost-sensor LED blink, Pico resets (USB re-enumeration) and stuck notes (MIDI monitor)	Arc, if any, to the GND track only (not across Rx); no reset, no lost sensor and no stuck note after any strike; any I^2C error recorded and reported (ADR-020)	REQ-023
V-26	MBE 0414 body and end-cap length (C2-002)	Caliper on one resistor before fitting: body length L and cap length L_c; then re-run `check_gap.py` section (d) with the measured values	$L_c \leq 3.0$ mm (4 kV rule met at every swept L and placement error, CALCULATION); otherwise the recomputed cap gap ≥ 3.5 mm at the measured placement error	REQ-023
V-27	I^2C traffic against electrode baseline noise (C2-012)	`SERIAL_DEBUG` deltas per electrode with the bus idle between reads versus continuous I^2C polling, electrodes on typical objects	No false Note On; noise with traffic increase of at most 3 counts in the per-electrode delta (ASSUMPTION threshold, T-074)	REQ-038
V-28	ADR-021 parts and pads	Test-fit C1 (5.08 mm leads) in its 0.8 mm holes; bridge JP1 with an R13–R16 lead off-cut over the 1.87 mm gap; tighten and re-tighten every J2 screw; with the pot fitted, log `analogRead` during slow and fast turns	C1 seated flat without force; JP1 bridge on one side only, no continuity to the other side; no J2 pad lifted or cracked; the octave follows the knob with no visible lag (settling ≤ 2.89 ms, CALCULATION)	REQ-006, 025, 029

11 Reproducibility

11.1 Toolchain

From [management/TOOLCHAIN.md] (MEASUREMENT by command, 2026-10-30) and the task handoffs:

- Host: macOS 26.5.2, arm64, Apple M4.
- KiCad 10.0.4 (kicad-cli and pcbnew Python 3.9.13).
- arduino-cli 1.5.1; arduino-pico core rp2040 6.3.1; AVR core 1.8.8; Adafruit MPR121 1.2.1, BusIO 1.17.4, MIDI Library 5.0.2, MIDIUSB 1.0.5.
- Python 3.9.6 with pytest 8.4.2 (project `.venv`); node v26.4.0; playwright-core 1.64.1 and Google Chrome 156.0.8078.74 (T-020).
- TinyTeX-1 v2026.10, pdfTeX 1.40.29 (TeX Live 2026), with siunitx, enumitem and caption added (documentation handoff T-006-DOC/T-031).
- MCP adapter `eng-tools` (`scripts/mcp/server.mjs`, @modelcontextprotocol/sdk 1.32.2).

11.2 How to rebuild

- Hardware: `sh hardware/scripts/build_all.sh` regenerates the schematic and PCB, runs ERC and DRC with schematic parity (fails on violations) and exports `hardware/fabrication/`.
- Firmware build matrix: eng-tools `firmware_compile`, or `arduino-cli compile` with the FQBN `rp2040:rp2040:<board>:usbstack=tinyusb` for rpipico, rpipicow, rpipico2, rpipico2w, each without and with `--build-property compiler.cpp.extra_flags=-DFRUTAMIDI_PCB`, plus `arduino:avr:leonardo` [management/INTERFACES.json].
- Host tests: eng-tools `run_pytest` with `tests/firmware` and `tests/host`.
- Systems checks: `sim/systems/budgets.py`, `interface_check.py`, `pcb_rules_check.py`; electronics calculations `sh sim/electronics/calc/run_all.sh`.
- Report: `sh reports/report/build.sh` (regenerates the requirements table from the CSV, then three pdflatex passes), or eng-tools `latex_build` with `reports/report/frutamidi_report.tex`.

Released file checksums: `releases/V1_BASELINE/SHA256SUMS` for the previous release (verify with `shasum -a 256 -c SHA256SUMS`). Release V1.2 (tag `v1.2.0`) has not been made; its checksum file will be created at release. A document cannot contain its own checksum, so this report's hash is listed there. The board figures in `reports/figures/` are byte-identical copies of `hardware/fabrication/images/` at a216c1b (sha256 in the T-078 handoff).

12 Change history

Baseline: release V1_BASELINE, git tag `v1.1.0-baseline` (048023c, 2026-11-04), board label "FrutaMIDI v1.1", interfaces frozen-3. Current: FrutaMIDI v1.2 on branch `change/CR-002` at f457bfe (manager PCB review follow-up, interfaces frozen-6, requirements draft-8), to be released as V1.2 (git tag `v1.2.0`) after user approval. Sources: `git log v1.1.0-baseline..f457bfe`, the commit messages quoted below, [changes/CR-002.md] ADR-019, ADR-020 and ADR-021 [management/DECISIONS.md].

Commit	Change	Effect on nets, copper and checks
21a4919	Schematic readability: A3 sheet, framed functional blocks, electrode chain drawn with wires. Layout of the schematic only.	Netlist identical (127 nets, compared node by node); PCB unchanged; ERC 0, DRC 0, parity 0 (commit message).
b1e4bf5	The unconnected BOM-only socket symbols J4–J7 are removed; the female-header sockets become a Socket field of A1/A2; the BOM gains a Socket column.	Netlist unchanged apart from the removed J4–J7 no-connect nets; PCB copper identical (tracks and pads compared); ERC 0, DRC 0, parity 0 (commit message).
130e754	CR-002 opened: electrode resistor flashover and pulse rating; supersedes candidate CR-001.	Documents only.
3a3684b	CR-002 T-060: R1–R12 become Vishay MBE04140C3301FC100 (DIN0414, lying flat, 15.24 mm pitch); relayout (module A2 15.1 mm up and 2.54 mm right; J1, JP1 upright, R13–R15 right of the module; H4 moved); analysis.md §5; ADR-019.	ELE–E gap 0.94 → 13.04 mm, E–other net 1.99 → 10.92 mm (MEASUREMENT, design file); nets and pinout unchanged; ERC 0, DRC 0, parity 0; pcb_rules_check 16/16.
8394b41	CR-002 T-070: REQUIREMENTS draft-6, INTERFACES frozen-4, budgets revision 6 (core BOM 23.78–37.72 EUR, CALCULATION), risks R-006 rewritten and R-008 added; board label “FrutaMIDI v1.2”.	Label text only on the board; gap check unchanged (re-run in T-072).
dcdb1fc	CR-002 T-072: this report, the user manual and figures updated for v1.2.	Documents only.
1247c44	CR-002 T-071: independent re-review findings C2-001 to C2-013 recorded (Section 8.1).	Documents only.
c901ad8	CR-002 T-073: review fixes. End-cap (cap-to-cap) and mounting-metal checks added to `check_gap.py`; pulse energy corrected to 182 mJ; R1–R12 drill 1.0 → 1.1 mm; H3 moved (95, 48) → (96, 38) mm; terminal labels moved off the pads; Ex–GND gap kept as a spark gap with a ground-bounce estimate (`calc_gnd_bounce.py`); assembly guide reworded (lead forming on a gauge, JP1 off-cut, nylon screw at H3, flat configuration wires); ADR-020.	Nets and pinout unchanged; copper gaps 13.04 / 10.92 mm unchanged; end caps 4 kV PARTIAL (worst 3.61 mm at ±0.25 mm); ERC 0, DRC 0, unconnected 0; pcb_rules_check 16/16; interface check 80/80 (MEASUREMENT / CALCULATION, T-073 handoff).
2ba289a	CR-002 T-074/T-075: REQUIREMENTS draft-7, INTERFACES frozen-5, risks R-009 and R-010; this report (CR-002 re-review section, ESD section, validation rows V-25 to V-27), README, user manual and figures updated after T-073.	Documents and check outputs only.
565ffb5	Report build fix: requirements text without U+222B; report rebuilt.	Documents only.

Commit	Change	Effect on nets, copper and checks
a216c1b	T-076, manager PCB review follow-up (ADR-021): JP1 pitch 2.54 → 3.81 mm; J2 pads 3.0 mm round → 3.0 × 4.5 mm oval, offset 0.75 mm towards the block back; RV1 mounting tabs on AGND; new C1 100 nF KEMET C322C104K5R5TA from the pot wiper to AGND; silkscreen overlaps removed; `calc_pot_rc.py` added; assembly guide updated.	New part C1 and new AGND members (RV1 tabs, C1 pin 2); other nets and pinout unchanged. JP1 gap 0.60 → 1.87 mm; J2 copper 7.1 → 11.6 mm^2 per pad; Ex–GND 1.40 mm unchanged; 0.8 mm holes 11 → 13, 133 holes in total; ERC 0, DRC 0, parity 0 (MEASUREMENT / CALCULATION, T-076 handoff).
f457bfe	T-077: interface check AGND membership rules (replacing `pcb.A1.AGND_isolated`), home-etch checks for the JP1 gap and J2 pads, budgets revision 7 (check A7, C1 row), INTERFACES frozen-6, REQUIREMENTS draft-8.	Checks only: interface check 85/85, pcb_rules_check 19/19, budgets 30 checks, 25 PASS, 5 FAIL; core BOM 23.98–38.62 EUR (CALCULATION).
(T-078)	This report (manager PCB review section, C1 and AGND, cost, interfaces), README, user manual and figures updated after T-077.	Documents only.

What did not change in v1.2. Firmware, pins and build flags; the J2 terminal row and pinout; the I^2C, USB MIDI and terminal interfaces apart from the R1–R12 part, C1, the AGND members and the JP1 and J2 footprints (ADR-021); the resistor value (3.3 kΩ 1 %) and therefore the electrode sensitivity analysis; the home-etch design rules (ADR-019 D-4, [`management/INTERFACES.json`]).

Requirement status after CR-002. Unchanged counts in REQUIREMENTS draft-8 at f457bfe (PASS 17, PARTIAL 18, FAIL 3, NOT RUN 4). REQ-023 stays PARTIAL (no ESD-gun test); REQ-006 stays PARTIAL (C1 settling by calculation only); REQ-032 stays FAIL with a larger overrun (8.62 EUR instead of 0.40 EUR, CALCULATION) [`management/REQUIREMENTS.md`].

A Sources

Source registers: [`management/SOURCES.md`] (S-0xx), [`requirements/SOURCES_SYSTEMS.md`] (S-xx) and [`reports/electronics/SOURCES_EE.md`] (EE-Sxx, marked "Read" or "Search"). The registers hold the full entries, access notes and checksums. Sources cited in this report:

ID	Source
S-01	Raspberry Pi Pico datasheet RP-008307-DS-2 (local copy, sha256 in the register).
S-02	Raspberry Pi documentation, "Pico-series microcontrollers", pin functions. `https://www.raspberrypi.com/documentation/microcontrollers/pico-series.html`
S-04, S-006, EE-S01	NXP/Freescale MPR121 datasheet, Rev. 4, 02/2013. `https://www.nxp.com/docs/en/data-sheet/MPR121.pdf`
S-06	Adafruit Eagle board files for the MPR121 breakout (STEMMA QT revision). `https://github.com/adafruit/Adafruit-MPR121-PCB`
S-11	Adafruit TinyUSB Arduino (in arduino-pico 6.3.1), device and MIDI sources.

ID	Source
S-14	USB Device Class Definition for MIDI Devices, Release 1.0 (1999-11-01). `https://www.usb.org/sites/default/files/midi10.pdf`
S-16	Raspberry Pi Pico 2 datasheet RP-008299-DS-3 (local copy).
S-17	Raspberry Pi RP2040 datasheet RP-008371-DS-1 (local copy).
EE-S08	Raspberry Pi Pico pinout. `https://pico.pinout.xyz/`
EE-S14b	Yageo leaded resistors datasheet (MFR series).
EE-S17	Vishay Beyschlag MBA/MBB/MBE professional thin-film resistors, doc 28766. `https://www.vishay.com/docs/28766/mbxsma.pdf`
EE-S17b	Same document, local copy (sha256 201a7ea8…9cff in the register); 1.2/50 µs pulse chart read from the vector paths: 0414 4.08 kV, 0207 1.74 kV, 0204 1.47 kV at 3.3 kΩ; 0414 minimum lead pitch 15.0 mm.
EE-S24	Vishay BCcomponents LSR37 high-voltage surge resistor, doc 28735 (third-party mirror, local copy): 12 kV ESD contact discharge rating, ±10 %/±20 %.
EE-S25	IEC 60664-1 Table F.2, clearance against rated impulse voltage. Values recalled, standard not opened: to be checked.
EE-S26	Distributor listings for sibling values of the MBE 0414 (RS, LCSC), basis of the 0.30–0.66 EUR per-part price; 3k3 price assumed equal.